NLP Chatbot

Industrial Safety

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1. Introduction

Natural Language Processing (NLP) is a field of artificial intelligence focused on enabling computers to understand, interpret, and generate human language. It bridges the gap between human language and computer understanding, allowing for more intuitive and effective interactions with technology.
Key aspects of NLP:

- Understanding human language: NLP algorithms analyses text and speech to extract meaning, sentiment, and intent.
- Processing and manipulating language: NLP can be used to translate languages, summarize text, generate new text, and more.
- Applications:
  NLP has a wide range of applications, including:
  - Speech recognition
  - Machine translation
  - Text summarization
  - **Chatbots and virtual assistants**
  - Sentiment analysis
  - Information extraction

1.1 Problem Statement

Safety remains a significant concern for industries worldwide, particularly in high-risk environments such as manufacturing, mining, and metal industries. Despite strict safety regulations, accidents continue to occur, often resulting in severe consequences. To address this issue, an AI-powered chatbot is designed to assist industrial safety professionals in identifying and mitigating workplace hazards.

The project is centered on improving industrial safety by developing a chatbot that employs Natural Language Processing (NLP). It aims to address the vital issue of analyzing and preventing accidents and injuries within industrial environments. The importance of this initiative is highlighted by the frequent occurrence of such incidents, which in some cases result in fatalities across industries worldwide. By leveraging Natural Language Processing (NLP) and Machine Learning (ML), the project seeks to strengthen safety measures and promote greater awareness in industrial settings. The system utilizes accident records from 12 manufacturing plants across three countries, incorporating data on accident severity, potential hazards, and critical risks.

The chatbot employs machine learning classifiers such as Random Forest and Boosting Classifiers to process accident descriptions with high accuracy and reliability. By integrating this tool into workplace safety frameworks, organizations can proactively identify risks, enhance regulatory compliance, and reduce the occurrence of industrial accidents. This study serves as a foundation for further refining and deploying the chatbot as an essential component of industrial safety management.

1.2 Data Description

The database comes from one of the biggest industries in Brazil and in the world. It is an urgent need for industries/companies around the globe to understand why employees still suffer some injuries/accidents in plants. Sometimes they also die in such an environment.

This database is basically records of accidents from 12 different plants in 03 different countries where every line in the data is an occurrence of an accident.
Columns description:

- Data: timestamp or time/date information
- Countries: which country the accident occurred (anonymised)
- Local: the city where the manufacturing plant is located (anonymised)
- Industry sector: which sector the plant belongs to
- Accident level: from I to VI, it registers how severe was the accident (I means not severe but VI means very severe)
- Potential Accident Level: Depending on the Accident Level, the database also registers how severe the accident could have been (due to other factors involved in the accident)
- Genre: if the person is male of female
- Employee or Third Party: if the injured person is an employee or a third party
- Critical Risk: some description of the risk involved in the accident
- Description: Detailed description of how the accident happened.

1.3 Key Findings

1.3.1 Data Insights and Trends:

The dataset reveals that workplace accidents are concentrated in specific industries and locations. Mining and metal industries account for most reported incidents, with 56.7% of accidents occurring in mining and 32.1% in metals. This indicates that these sectors require enhanced safety measures and stricter compliance protocols.
Accident severity distribution shows that 73.9% of incidents are minor (Level I), while more severe accidents (Levels IV and V) occur less frequently. However, potential accident levels suggest that many incidents could have escalated in severity, emphasizing the need for better risk mitigation strategies.
From a geographical perspective, Country_01 records the highest number of accidents (59.3%), followed by Country_02 (30.9%). Local_03 emerges as a high-risk location with the most reported accidents, suggesting that specific plant locations require additional safety interventions.

1.3.2 Workforce and Safety Considerations

An analysis of workforce demographics reveals that 44.3% of accidents involve third-party workers, indicating a potential gap in safety training and compliance for contractors. Ensuring that third-party employees receive the same level of safety training as full-time employees is essential to reducing workplace risks.
The data also highlights a significant gender disparity, with 94.7% of accidents involving male workers and only 5.3% affecting female workers. While this may reflect the

workforce composition of the industry, it suggests the need for inclusive safety policies and targeted safety measures for all employees.

1.3.3 Critical Risk Factors Identified

- Pressed Incidents: Many accidents involve workers being pressed between objects or machinery, indicating the need for automated pressure release mechanisms and stricter machine operation protocols.
- Chemical Substances and Gas Release: Exposure to hazardous substances remains a significant risk, emphasizing the importance of regular inspections, protective equipment, and emergency response training.
- Manual Tool-Related Incidents: Many accidents result from improper tool use or equipment failure, reinforcing the need for ergonomic tool design, hands-on safety training, and stricter equipment maintenance procedures.

Imbalanced Distribution of Accident Levels

The dataset shows a high imbalance in accident severity levels, with Level I (low severity) accidents accounting for 73.9% of all incidents, while severe accidents (Levels IV and V) occur much less frequently. This imbalance suggests that while most reported incidents are minor, the less frequent severe accidents still pose a significant threat to workplace safety and require focused intervention

2. Data Collection and Preprocessing

The foundation of any machine learning project lies in acquiring and preparing high-quality data. In our project, accident records were collected from 12 manufacturing plants across three countries, detailing factors such as severity, potential hazards, and critical risks.

Steps taken to achieve this:

- Loading of Data
- Cleaning of Data
- Standardizing the Columns

Detailed description for each step will be explained as we proceed

2.1 Loading Data Set

The dataset was imported and an initial check with df.info () provided an overview of the dataset, including data types and missing values

The dataset contains 425 entries with 10 columns

	Index	Data	Countries	Local	Industry Sector	Accident Level	Potential Accident Level	Genre	Employee or Third Party	Critical Risk	Description
0	0	1/1/16 0:00	Country_01	Local_01	Mining	I	IV	Male	Third Party	Pressed	While removing the drill rod of the Jumbo 08 f...
1	1	1/2/16 0:00	Country_02	Local_02	Mining	I	IV	Male	Employee	Pressurized Systems	During the activation of a sodium sulphide pum...
2	2	1/6/16 0:00	Country_01	Local_03	Mining	I	III	Male	Third Party (Remote)	Manual Tools	In the sub-station MILPO located at level +170...
3	3	1/8/16 0:00	Country_01	Local_04	Mining	I	I	Male	Third Party	Others	Being 9:45 am. approximately in the Nv. 1880 C...
4	4	1/10/16 0:00	Country_01	Local_04	Mining	IV	IV	Male	Third Party	Others	Approximately at 11:45 a.m. in circumstances t...

Performed following steps for preprocessing the data before carrying out EDA

- 'Index' is just an indexer column having no significance in data analysis. We would go ahead and remove the column from our set.
- We removed 7 duplicate rows from the dataset.
- Confirming our initial analysis, we see there are no null values.
- We then renamed the columns to make it a more meaningful column.

Following is the cleaned dataset which we have created after performing above steps.

2.2 Cleaned Dataset

```
cleaned_data.head()
```

	Date	Countries	Local	Industry_Sector	Accident_Level	Potential_Accident_Level	Gender	Employee_or_Third_Party	Critical_Risk	Description	Year	Month	DayOfWeek	Quarter
0	2016-01-01	Country_01	Local_01	Mining	I	IV	Male	Third Party	pressed	while removing the drill rod of the jumbo 08 f...	2016	1	Friday	1
1	2016-01-02	Country_02	Local_02	Mining	I	IV	Male	Employee	pressurized systems	during the activation of a sodium sulphide pum...	2016	1	Saturday	1
2	2016-01-06	Country_01	Local_03	Mining	I	III	Male	Third Party (Remote)	manual tools	in the sub-station milpo located at level +170...	2016	1	Wednesday	1
3	2016-01-08	Country_01	Local_04	Mining	I	I	Male	Third Party	others	being 9:45 am. approximately in the nv. 1880 c...	2016	1	Friday	1
4	2016-01-10	Country_01	Local_04	Mining	IV	IV	Male	Third Party	others	approximately at 11:45 a.m. in circumstances t...	2016	1	Sunday	1

Initial Data Observations

- There are total of Three countries and Twelve plants/cities overall,
- Five type of accident levels (1 to 5) are present.
- Six type of Potential Accident Levels (1 to 6) and we see that there is only one value registered under 'Accident level 6' and it can be **replaced to level 5** to reduce the complexity. Level VI accident is having only one incident hence it can be replaced with Level V as it is closer one
- For Feature selection we have removed unnamed column since it had no significance.
- Data manipulation –
  - split date column into day, month, year columns,
  - one-hot encoding
  - In the Potential_accident_level column we have dropped level 6 as it only contained one observation.
- Dimensionality reduction has been for 2 Machine Learning algorithms - Naive Bayes and SVC

2.3 Standardizing Column names

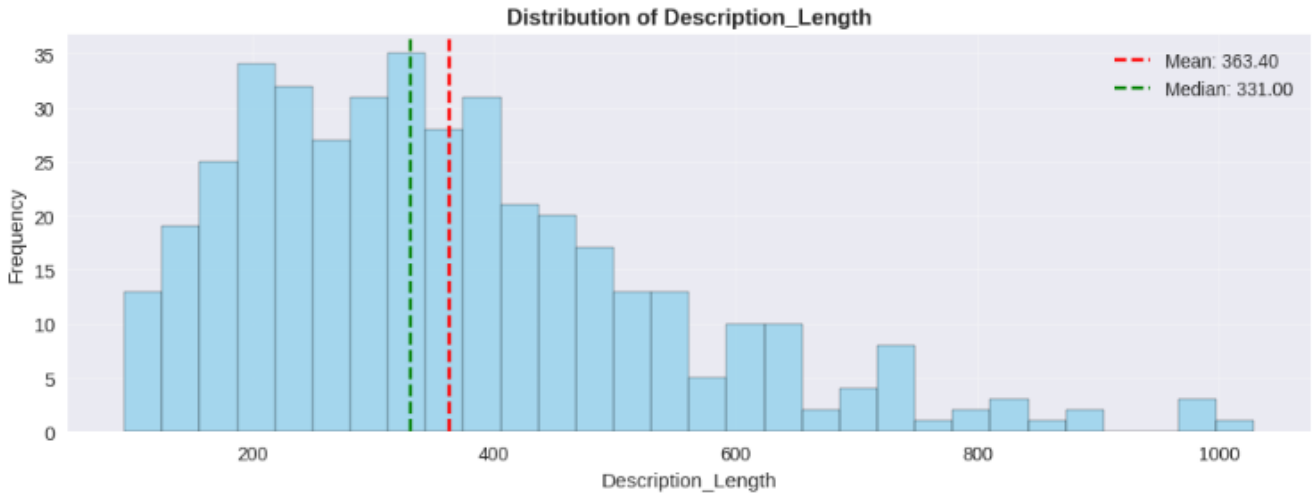
Columns seemed to be improperly named with no significant meaning from the title. It is necessary to rename the columns to understandable terms. The following columns were renamed:

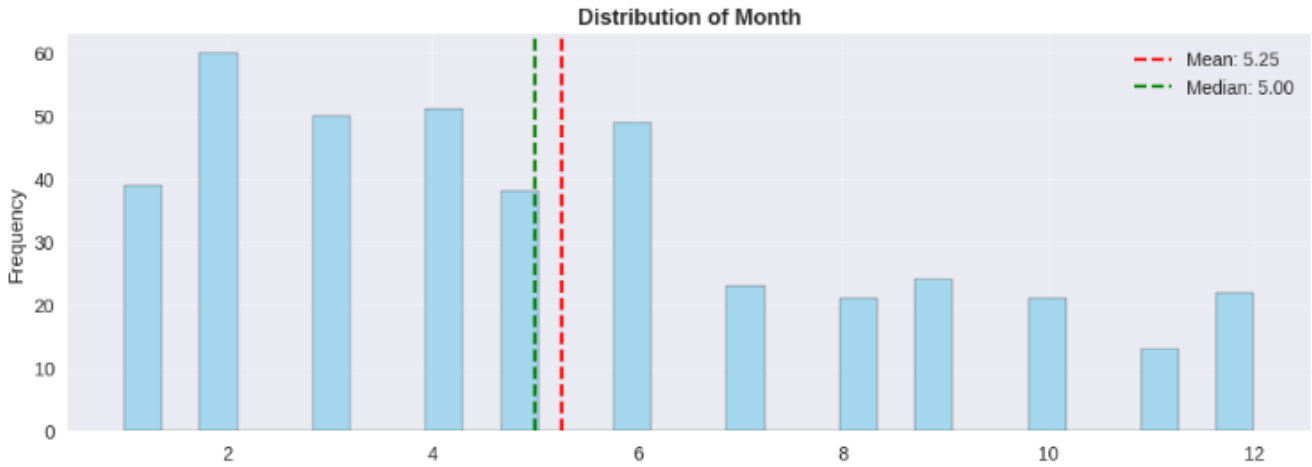
Original Dataset Column Name	**Renamed Column**
Data	Date
Countries	Country
Genre	Gender
Employee or Third Party	Employee type

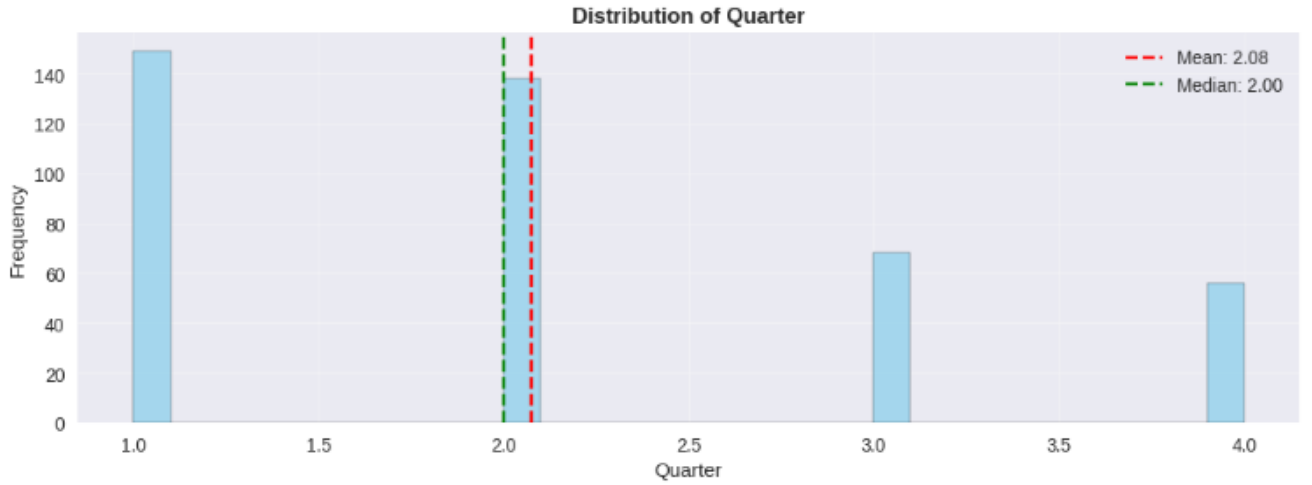
3. Exploratory Data Analysis

Exploratory Data Analysis (EDA) is a crucial step in understanding the dataset, identifying patterns, and preparing data for model development. EDA for the industrial safety chatbot involves the following:

3.1 Univariate Analysis



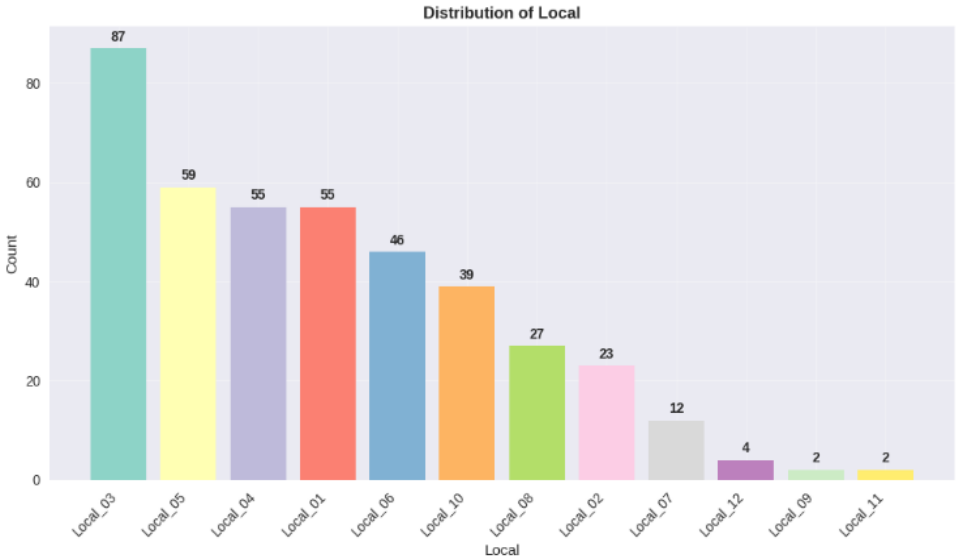




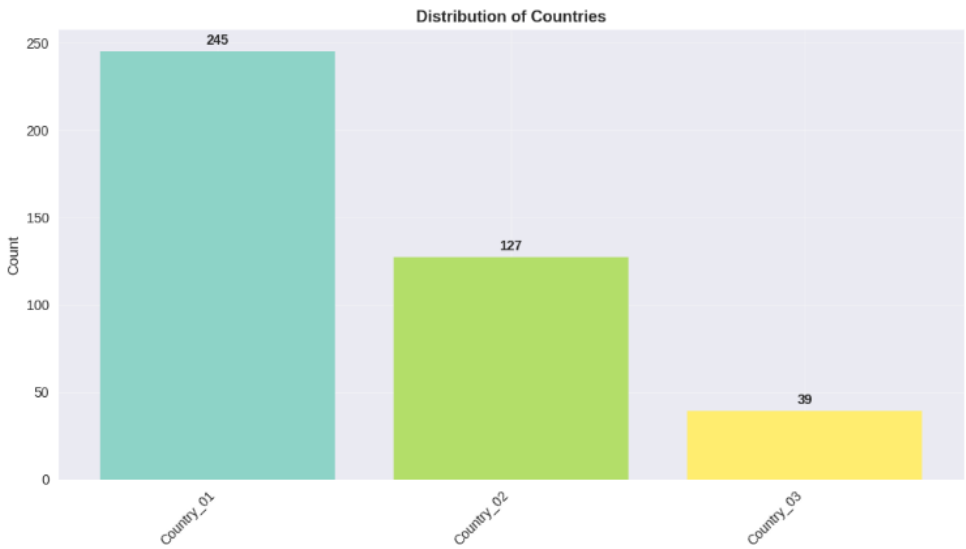
- Description length ranges from 100 to 1000 words and is a right skewed data.

- If we see on a high level, most of the accidents have taken place in the first half year and has gradually decreased in the second half.

Categorical Variables

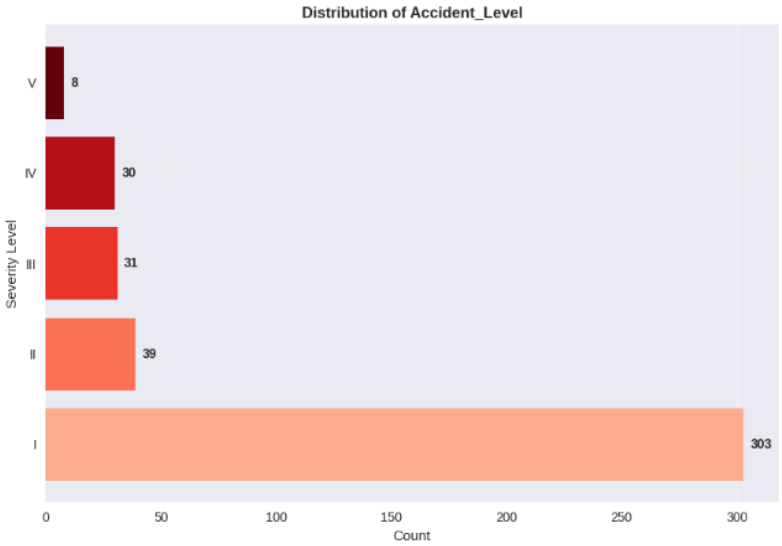


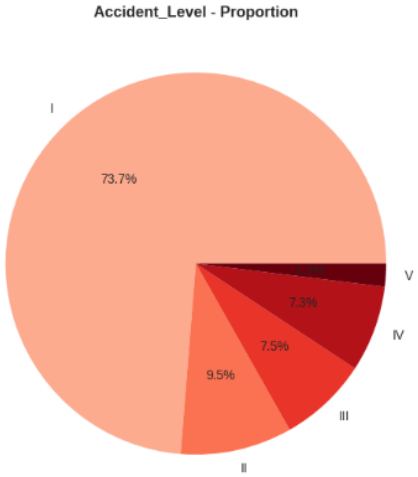
- Local 03 has the highest count of accidents
- Followed by there are local_05 and Local _04 with the second highest and third highest areas prone to accidents.
- Local 9 and 11 are least prone to accidents.

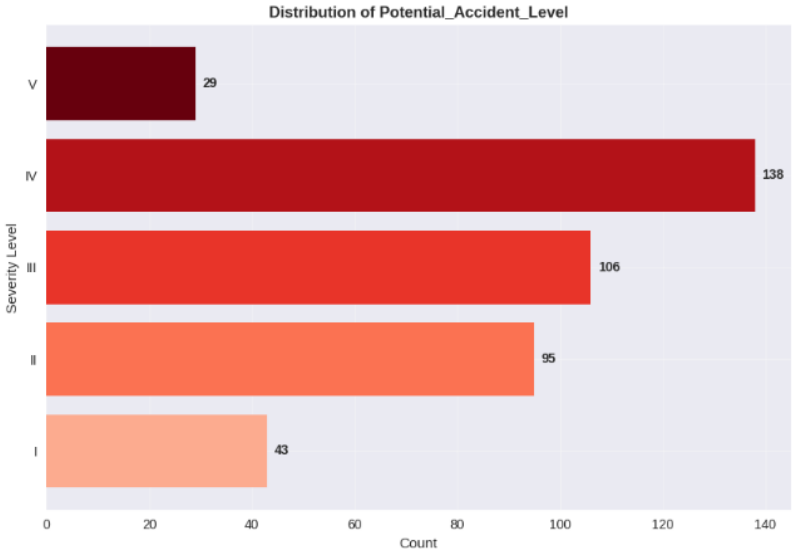


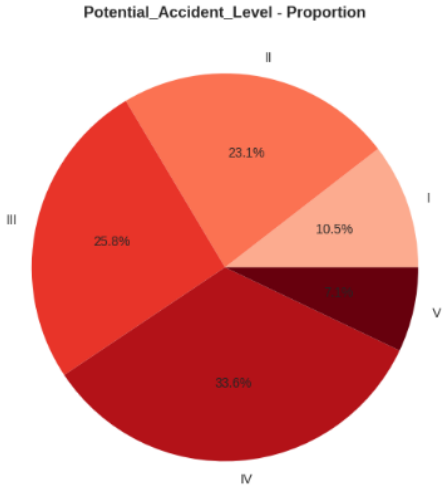
- Highest number of accidents have taken place in Country_01
- Country_03 has the lowest number of accidents.

Ordinal Variables:

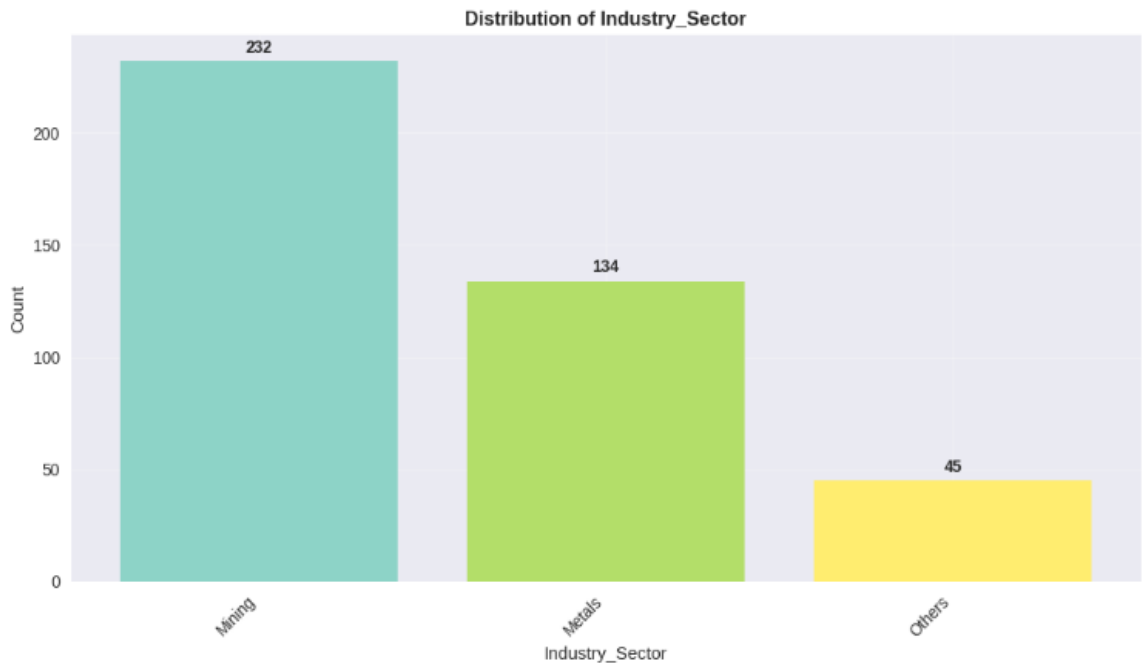




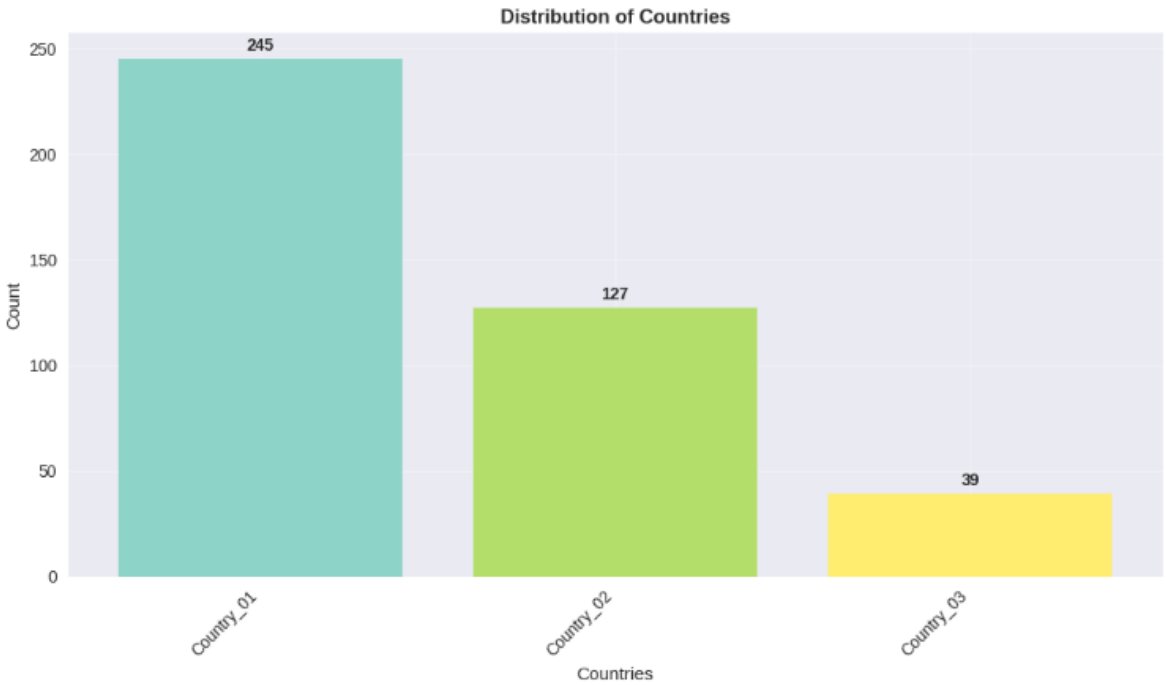




- Count of accidents is the highest for level 1 while the count for potential accident level is the highest for level 4.
- Level 5 is the least accident level for both Potential_accident_level and actual Accident level

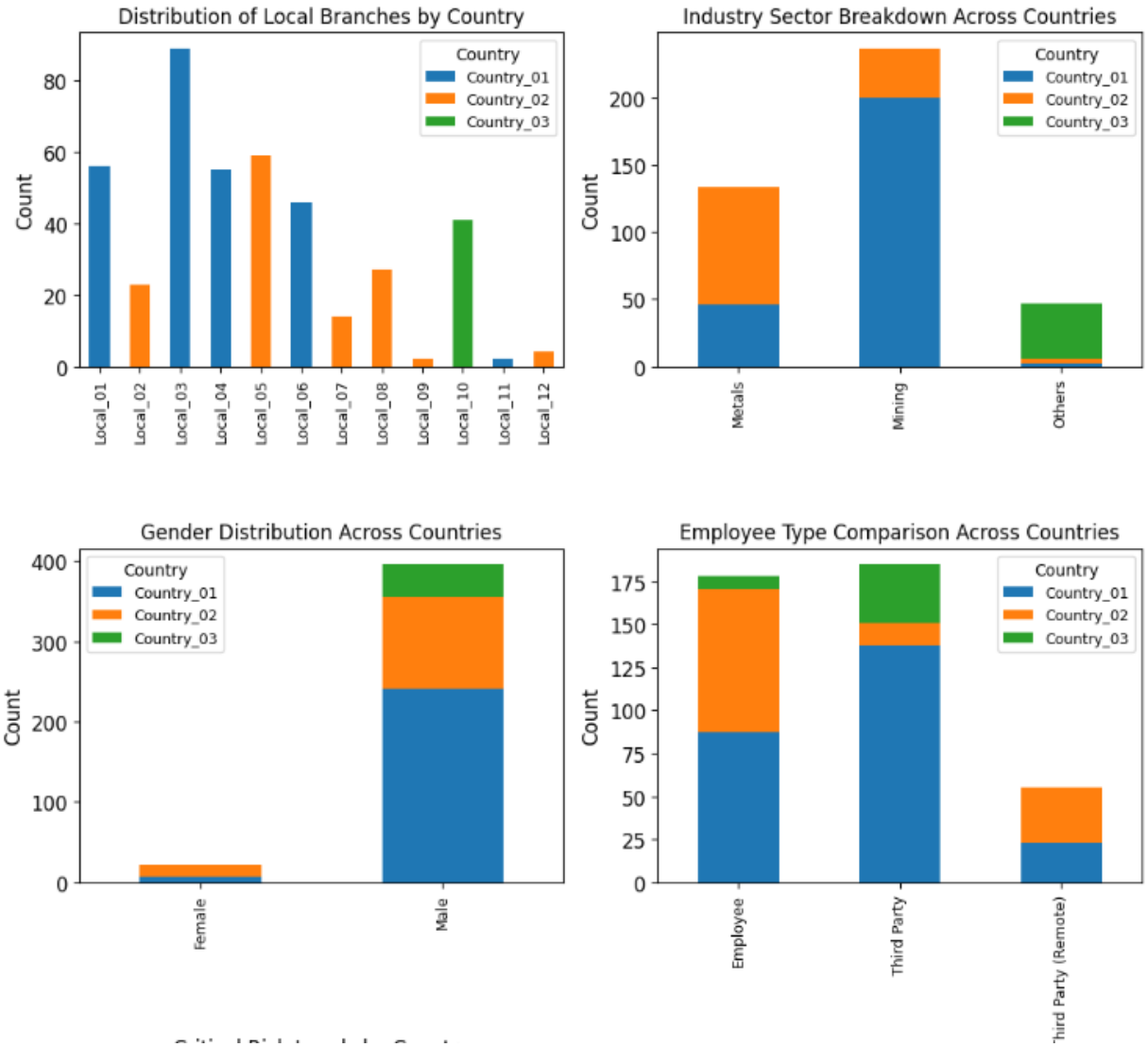


- Mining has the highest percentage at approximately 55%.
- Metals follow with around 30%.
- others have the lowest percentage, close to 15%.



- Country_01 has the highest percentage at approximately 58%.
- Country_02 follows with around 35%.
- Country_03 has the lowest percentage, close to 10%.

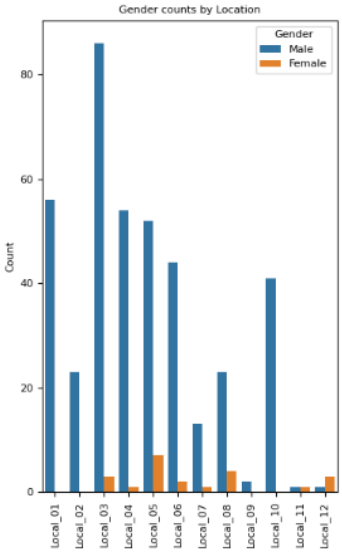
3.2 Bivariate Analysis

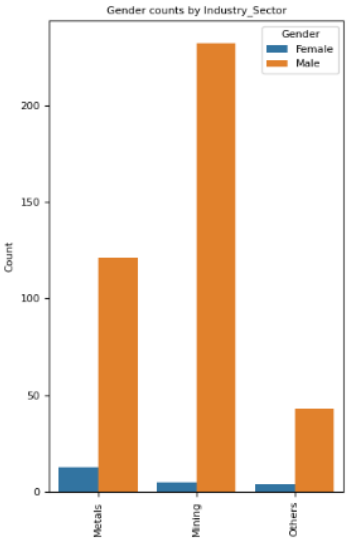


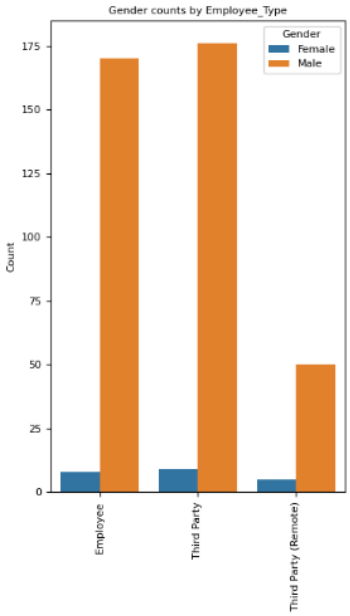
- Location Distribution
  Country_01 dominates Local_03, followed by Local_01 and Local_04. Country_02 is prominent in Local_05 and Local_08. Country_03 is concentrated in Local_10. Local_02, Local_06, Local_07, Local_09, Local_11, and Local_12 have low counts across all countries.
- Industry Sector
  Mining has the highest count, led by Country_01, then Country_02. Metals are significant in Country_01 and Country_02, with Country_02 slightly ahead. Other industries have minimal representation, with only Country_03 contributing.
- Gender Distribution
  Males dominate across all countries, with Country_01 having the highest count. Females are underrepresented, with only Country_02 contributing a small count.
- Employee Type
  Employees & Third-Party Workers: Country_01 leads, followed by Country_02 and

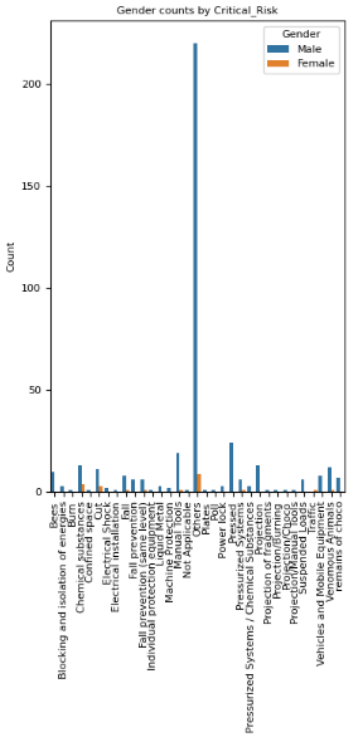
Country_03. Third-Party (Remote): Country_02 ranks highest, Country_01 follows, while Country_03 has none.

- Critical Risks
  "Others" category has the highest risk, mostly from Country_01. Key risks ("Pressed," "Manual Tools," "Cut," "Chemical Substances") are most common in Country_02. Other risks have low counts across all countries.

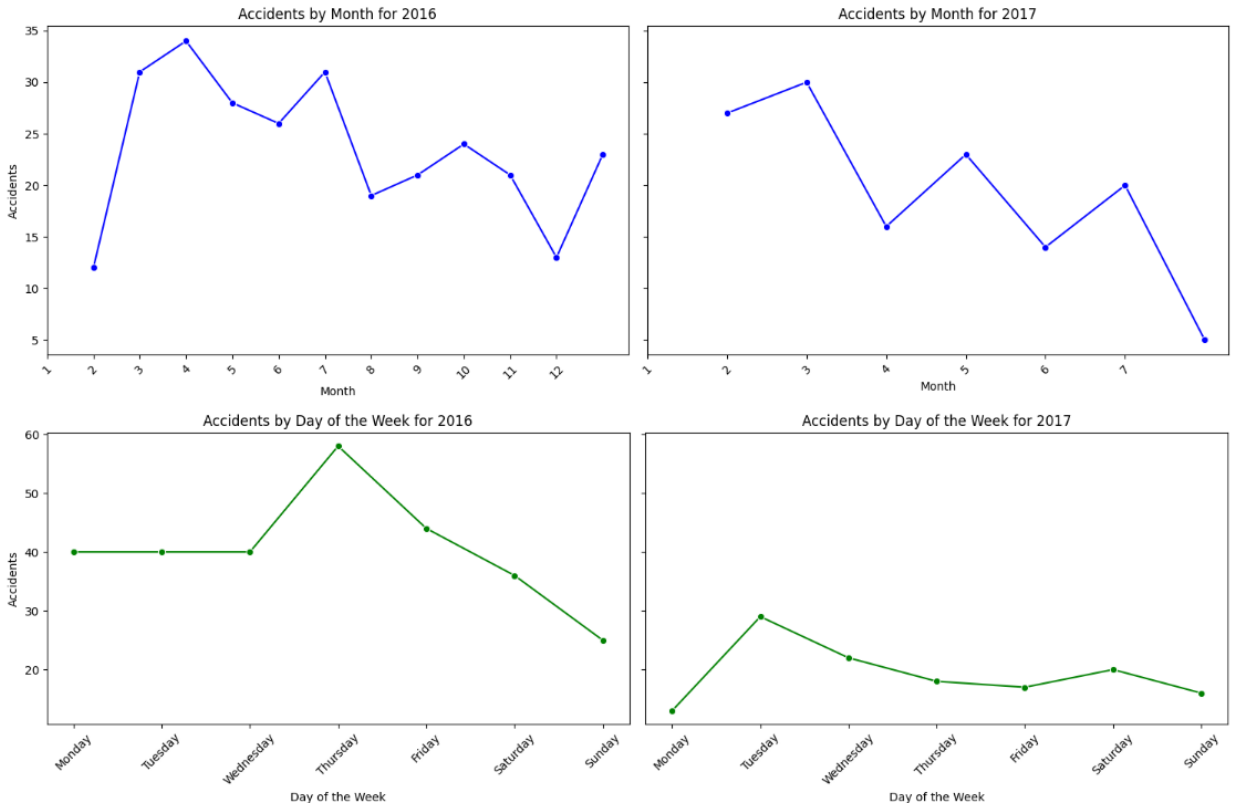








- Gender Counts by Local
  Local_03 has the highest count of males, followed by Local_01 and Local_04 Female counts are significantly lower across all locals, with Local_05 having the highest female count among them Males outnumber females in most of the locals.
- Gender Counts by Industry Sector
  The Mining sector has the highest count of males, followed by the Metals sector The Others sector has a noticeable count of females, but males still dominate in numbers
- Gender Counts by Employee Type
  Males dominate in all three categories: Employees, Third Party, and Third Party (Remote) The highest male counts are in the Employee and Third-Party categories Female counts are significantly lower in comparison to males in all categories
- Gender Counts by Critical Risk
  The Others category has the highest count of males by a large margin Other critical risk categories have relatively low counts, with males being more numerous than females in most categories
- These observations highlight gender distribution trends across different organizational and operational contexts



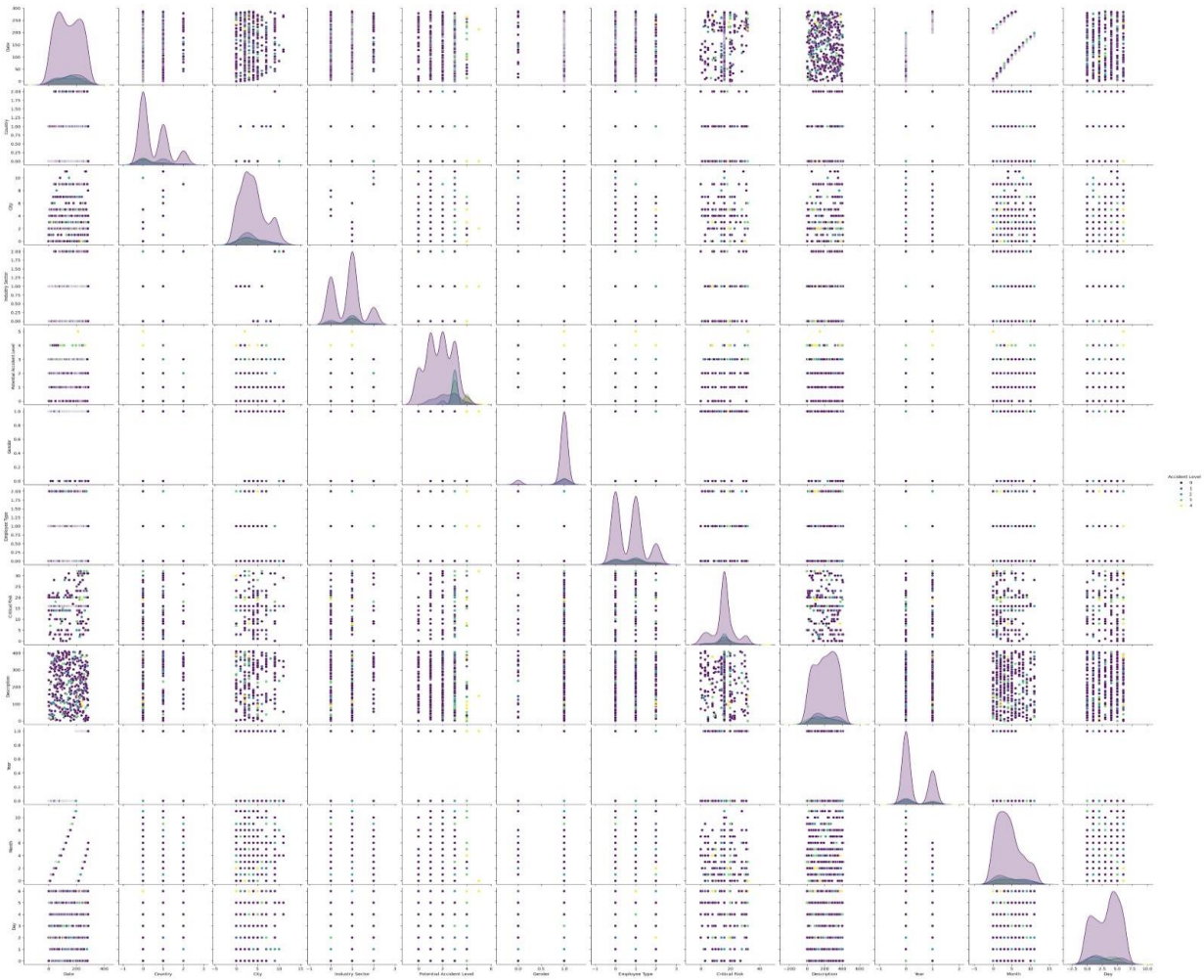
- Plot: Number of Accidents by Month - 2016
  Monthly Variability: The number of accidents fluctuates significantly throughout the year. Peak Month: March sees the highest number of accidents with around 34 incidents. Lowest Month: January records the lowest number of accidents, with around 12 incidents.
- Plot: Number of Accidents by Month - 2017
  Downward Trend: A consistent decrease in the number of accidents is observed over the months. Peak Month: February has the highest number of accidents, with around 30 incidents. Lowest Month: July records the lowest number of accidents, with about 5 incidents.
- Number of Accidents by Day of the Week - 2016
  Steady Weekdays: Accidents are relatively stable from Monday to Wednesday, with around 40 incidents each day. Thursday Spike: There's a significant increase on Thursday, reaching approximately 60 accidents. Weekend Decline: After Thursday, accidents decrease, with Friday having around 40, Saturday around 30, and Sunday dropping to the lowest at around 20 accidents.
- Number of Accidents by Day of the Week - 2017
  Starting Low: Monday starts with less than 10 accidents. Tuesday Peak: A sharp increase to about 30 accidents on Tuesday. Gradual Decline: From Wednesday to Friday, accidents decrease gradually (Wednesday ~25, Thursday ~20, Friday ~15). Weekend Variation: A slight increase on Saturday (around 20 accidents), followed by a drop on Sunday to around 10 accidents.

Correlation Insights

- Strong Correlations
  Year and Date (0.81): This is expected, as the year is derived directly from the date. Country and City (0.71): A strong correlation indicates that cities are grouped under specific countries, which is logical.
- Moderate Correlations
  Accident Level and Potential Accident Level (0.51): Suggests that the severity of an accident is somewhat related to its potential risk level. Critical Risk and Year (0.24): Shows a weak to moderate correlation, indicating trends over time related to critical risks.
- Negative Correlations
  Potential Accident Level and Country (-0.38): Indicates that different countries might report potential accident levels differently, possibly due to varying safety standards. Month and Year (-0.42): Likely due to variations in reporting frequency across different years and months.
- Weak or No Correlation
  Variables such as Gender, Employee Type, and Description show very weak correlations with other attributes, indicating little to no direct relationship. These insights help identify key relationships and potential areas for further analysis.

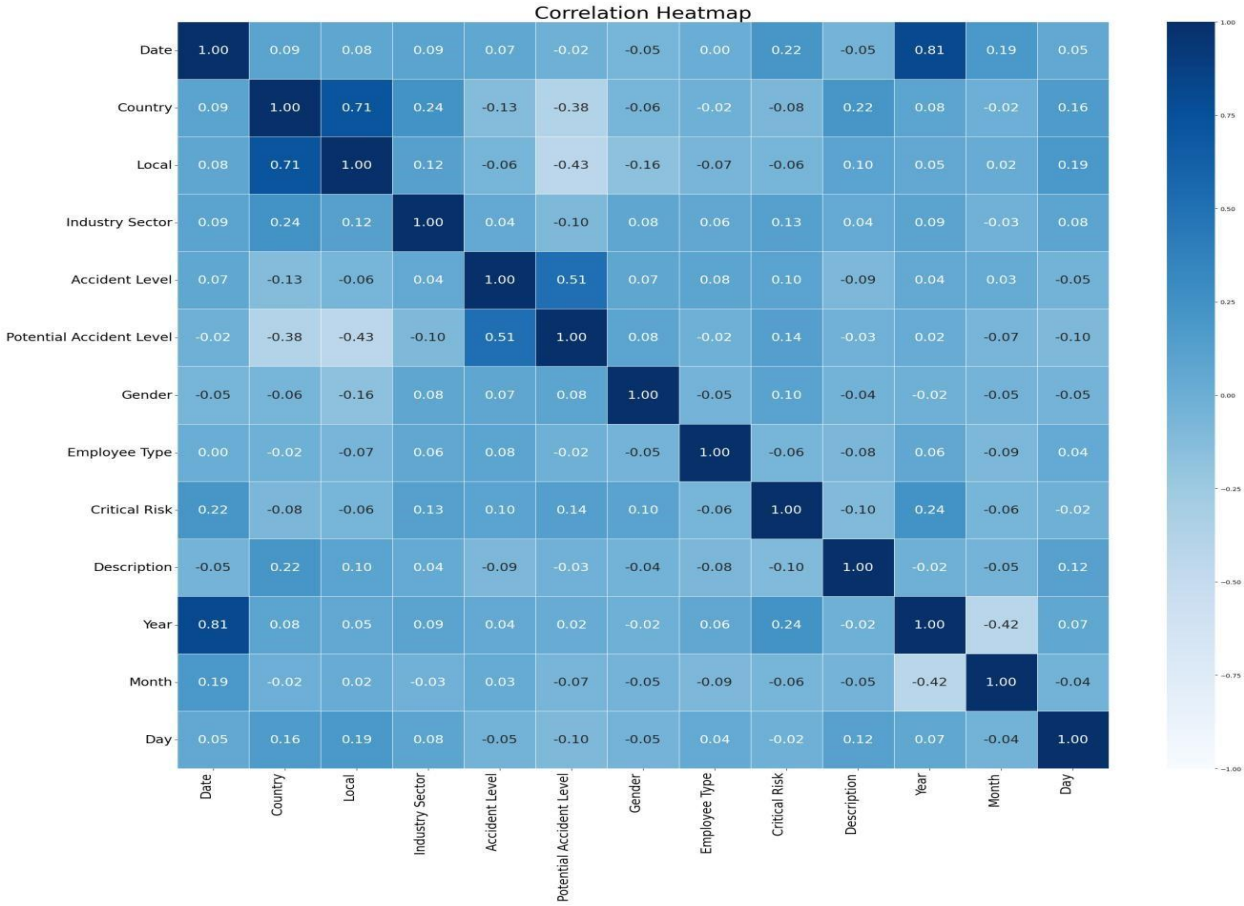
3.3 Multivariate Analysis

Pair Plot:

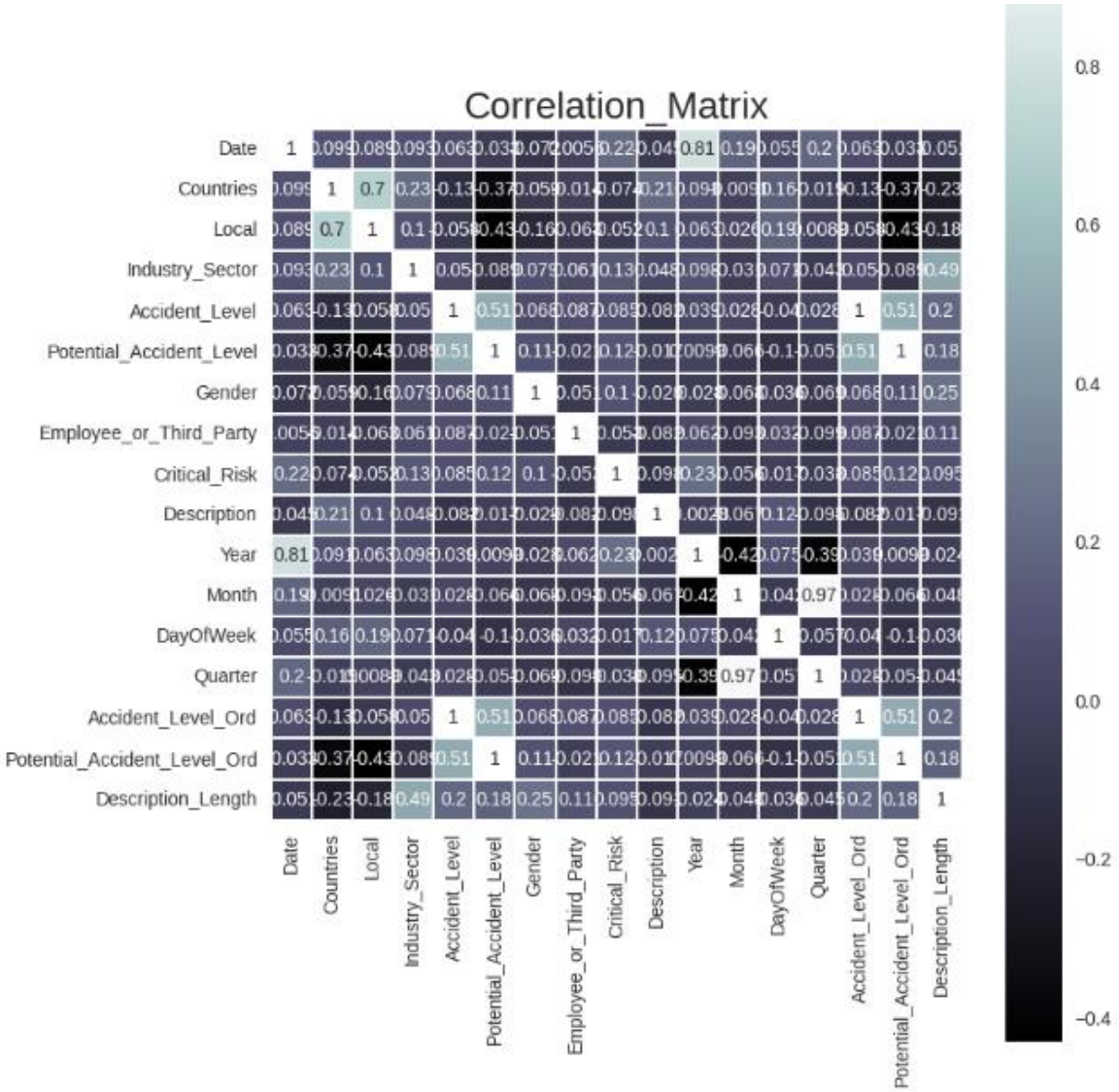


- Potential Accident Level: This is the most distinguishable factor in the pair plot concerning "Accident Level," making sense as it represents the estimated severity of an accident before it occurs.
- Employee Type: The pair plot shows some differentiation in accident severity based on the type of worker involved, though the impact is relatively subtle.
- Critical Risk: Certain critical risks appear more frequently in High Severity accidents when viewed in the pair plot, although the distinction is not very pronounced.
- Gender: The pair plot suggests that gender might have a small influence on accident levels, but this effect is not strong.
- Industry Sector: The industry type shows a weak pattern in the pair plot with accident levels, indicating that accidents happen across industries somewhat evenly.
- Local & Country: The geographical location has a slight negative pattern in the pair plot, suggesting that accident severity doesn't significantly depend on location.

Heat Map:



- **City & Country are closely linked (0.71 correlation)** – This makes sense since cities belong to countries! But it also hints that location-based trends impact incidents significantly.
- **Potential & Actual Accident Levels are moderately related (0.51 correlation)** – High-risk incidents don't always escalate, but when they do, the consequences can be serious. Risk management must focus on preventing escalation.
- **Year & Date have a strong bond (0.81 correlation)** – Expected, since dates include years! However, it also shows that trends might shift over time, requiring year-over-year analysis for better safety improvements.
- **Country weakly influences Industry Sector (0.24 correlation)** – Different industries pose different risks, but accidents don't seem heavily industry-dependent. This suggests safety measures are key across all sectors.
- **Month & Year have a negative correlation (-0.42)** – This could mean seasonal patterns in incident reporting. Certain months may have higher or lower accident rates due to operational or environmental factors.
- **Gender barely impacts accident levels (0.071 correlation)** – Safety risks don't discriminate! Regardless of gender, workplace hazards affect everyone, reinforcing the need for inclusive and comprehensive safety policies.



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- Employee Type: The pair plot shows some differentiation in accident severity based on the type of worker involved, though the impact is relatively subtle
- Critical Risk: Certain critical risks appear more frequently in severe accidents when viewed in the pair plot, although the distinction is not very pronounced
- Gender: The pair plot suggests that gender might have a small influence on accident levels, but this effect is not strong
- Industry Sector: The industry type shows a weak pattern in the pair plot with accident levels, indicating that accidents happen across industries somewhat evenly
- Local & Country: The geographical location has a slight negative pattern in the pair plot, suggesting that accident severity doesn't significantly depend on location

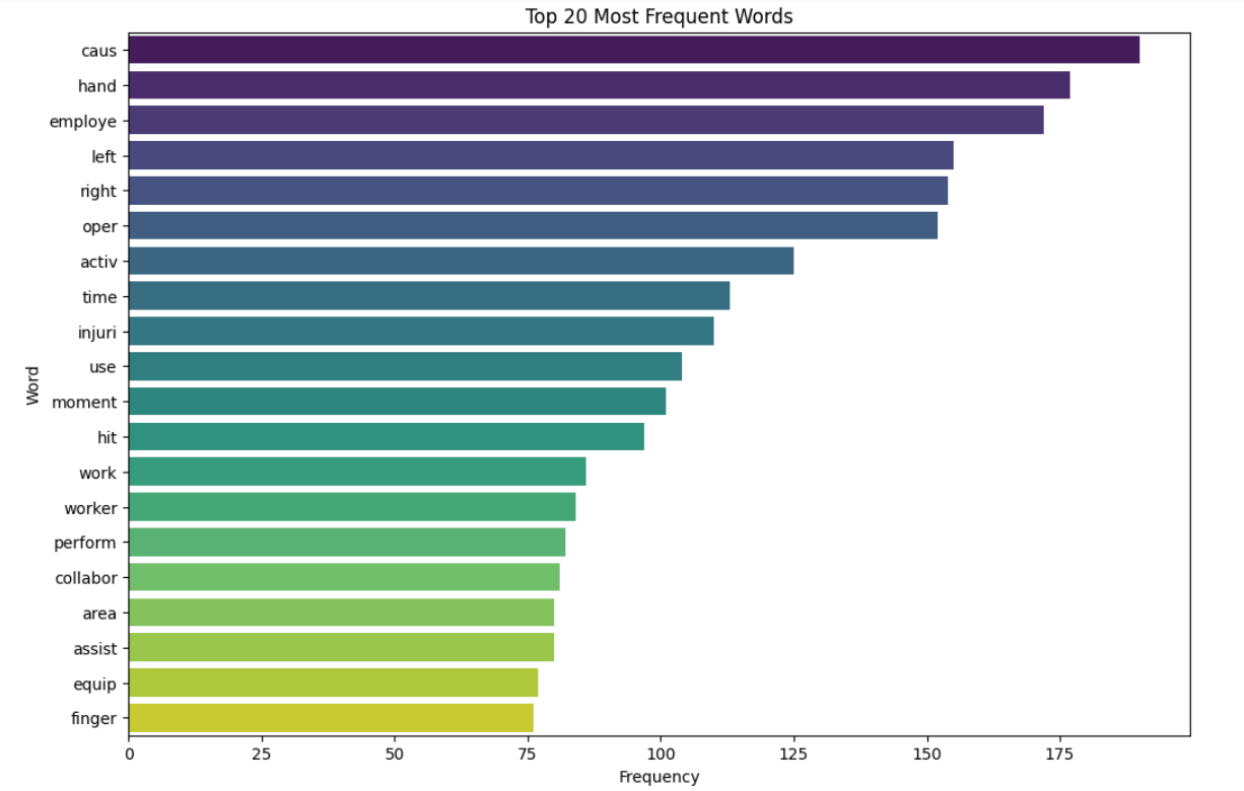
4. Feature Engineering and Embedding

4.1 NLP Preprocessing Cleanup

- Removing special characters
- Deleting Stop words
- Removed extra space
- Performed stemming

Generated Word cloud - Most frequent words used in the Description column.

Following are the top 20 most frequently used words:



4.2 Text Representation Algorithm

After that for representing the test column in vectors we have used the following algorithm

- WordToVec
- GloVe
- Sentence Transformer
- **TF-IDF**
- BOW

Dataset generated from each of the above algorithms was then split into training, validation and test dataset. Following table shows the size of each generated dataset.

Train, Validation, and Test Set Summary:

	Embedding Type	Training Samples	Validation Samples	Testing Samples	Training Shape	Validation Shape	Testing Shape
0	Word2Vec	250	84	84	(250, 200)	(84, 200)	(84, 200)
1	GloVe	250	84	84	(250, 100)	(84, 100)	(84, 100)
2	Sentence Transformer	250	84	84	(250, 384)	(84, 384)	(84, 384)
3	TF-IDF	250	84	84	(250, 2272)	(84, 2272)	(84, 2272)
4	BoW	250	84	84	(250, 1)	(84, 1)	(84, 1)

- **BoW:** Excellent for tasks where the presence or absence of keywords is sufficient. It's simple and fast.
- **TF-IDF:** Better than BoW for keyword importance, as it downplays common words and highlights unique ones.
- **Sentence Transformers:** These models generate dense embeddings that capture the meaning and context of entire sentences or even paragraphs.
- **GloVe**: Constructs a global word-word co-occurrence matrix for the entire corpus, which records how frequently words appear together within a specified context window.
- **WordToVec**: It's a "predictive" model that learns word embeddings by trying to predict context words given a target word, or vice-versa. It focuses on the local context of words.

Our models have outperformed on **TF-IDF** based dataset and hence we have used TF-IDF for calculating the test accuracy metrics.

4.3 Handling Class Imbalance and Encoding

Our data set originally categorizes accidents from Level I (least severe) to Level V (most severe) which we can see based on the description of categories:

- Level I: Minor Accident
- Level II: Slight Injury
- Level III: Moderate Injury
- Level IV: Severe Injury
- Level V: Very Severe Injury

- Now, since the dataset was very imbalanced, we have used SMOTE (Synthetic Minority Oversampling Technique) for balancing out the target classes. We have considered **Potential_Accident_Level** as our dependent variable(**Y**).
- Performed one hot encoding to convert categorical variables into numeric form.
- We then created a csv file output of this transformed data so that everyone from our group can train their models without any discrepancy.

5. Model Building

5.1 Model Selection

Logistic Regression algorithm:

Logistic regression is Machine Learning algorithms which comes under the Supervised Learning technique. It is used for predicting the categorical dependent variable using a given set of independent variables.
Logistic Regression can be used to classify the observations using different types of data and can easily determine the most effective variables used for the classification.

Naive Bayes Classifier algorithm:

Naive Bayes algorithm is a supervised learning algorithm which is based on Bayes theorem and used for solving classification problems.
It is mainly used in text classification that includes a high-dimensional training dataset. It is a probabilistic classifier which means it predict based on the probability of an object.
Some popular examples of Naive Bayes Algorithm are spam filtration, Sentimental analysis, and classifying articles.

Random Forest algorithm:

Random Forest is machine learning algorithm that comes under supervised learning technique. It is based on the concept of ensemble learning, which is a process of combining multiple classifiers to solve a complex problem and to improve the performance of the model.
Random Forest is a classifier that contains several decision trees on various subsets of the given dataset and takes the average to improve the predictive accuracy of that dataset.
The greater number of trees in the forest leads to higher accuracy and prevents the problem of overfitting.

Support Vector Machine (SVM) algorithm:

The goal of the SVM algorithm is to create the best line or decision boundary that can segregate n-dimensional space into classes so that we can easily put the new data point in the correct category in the future. This best decision boundary is called a hyperplane.
SVM chooses the extreme points/vectors that help in creating the hyperplane. These extreme cases are called as support vectors, and hence algorithm is termed as Support Vector Machine.

K-Nearest Neighbor (KNN) algorithm:

K-Nearest Neighbor is Machine Learning algorithms based on Supervised Learning technique.

K-NN algorithm stores all the available data and classifies a new data point based on the similarity. This means when new data appears then it can be easily classified into a well suite category by using K-NN algorithm.
K-NN is a non-parametric algorithm, which means it does not make any assumption on underlying data.
KNN algorithm at the training phase just stores the dataset and when it gets new data, then it classifies that data into a category that is much like the new data.

Decision Tree

Hierarchical Structure: A decision tree is a flowchart-like structure where each internal node represents a test on an attribute, each branch represents the outcome of the test, and each leaf node represents a class label (for classification) or a numerical value (for regression). It effectively partitions the data based on a series of questions.
Interpretability and Non-parametric: Decision trees are highly interpretable and easy to visualize, often mirroring human decision-making processes. They are also non-parametric, meaning they do not assume a specific underlying distribution of the data, making them versatile for various data types.

Bagging (Bootstrap Aggregating)

Parallel Training with Bootstrapping: Bagging involves creating multiple subsets of the original training data by bootstrapping (randomly sampling with replacement). Then, independent "weak learners" (often decision trees) trained in parallel on each of these subsets.
Variance Reduction and Overfitting Mitigation: The primary goal of bagging is to reduce the variance of a model, especially for high-variance, low-bias models like decision trees. By averaging or taking a majority vote of the predictions from multiple independently trained models, bagging helps to mitigate overfitting and improve the overall stability and accuracy of the model.

Boosting

Sequential Training with Error Correction: Boosting trains multiple "weak learners" sequentially. Each subsequent model built to correct the errors (misclassifications or high residuals) made by the previous models. This achieved by giving higher weights to the data points that misclassified or poorly predicted in the previous iteration.
Bias Reduction and Improved Accuracy: The core idea of boosting is to convert a set of weak learners into a single strong learner. By iteratively focusing on the "hard" examples, boosting aims to reduce the bias of the overall model and significantly improve its predictive accuracy, often outperforming individual models.

5.2 Model Training

We have first trained all the models namely

```python
class BasicMLClassifiers:
    """Comprehensive basic machine learning classifier implementation"""

    def __init__(self):
        self.classifiers = {}
        self.trained_models = {}
        self.results = {}

    def initialize_classifiers(self):
        """Initialize all basic classifiers"""

        self.classifiers = {
            'Random Forest': RandomForestClassifier(
                n_estimators=100,
                max_depth=10,
                random_state=42,
                n_jobs=-1
            ),
            'Logistic Regression': LogisticRegression(
                random_state=42,
                max_iter=1000,
                solver='liblinear'
            ),
            'SVM': SVC(
                random_state=42
            ),
            'Naive Bayes': MultinomialNB(
                alpha=1.0
            ),
            'Decision Tree': DecisionTreeClassifier(
                max_depth=10,
                random_state=42,
                min_samples_split=5
            ),
            'K-Nearest Neighbors': KNeighborsClassifier(
                n_neighbors=5,
                weights='distance'
            ),
            'Bagging': BaggingClassifier(n_estimators=50, max_samples=.7),
            'AdaBoost': AdaBoostClassifier(n_estimators= 50),
            'Gradient Boost': GradientBoostingClassifier(n_estimators = 50, learning_rate = 0.05),
            'XGBoost': XGBClassifier()
        }

        print("Initialized classifiers:")
        for name in self.classifiers.keys():
            print(f"  - {name}")
```

```
Initialized classifiers:
  - Random Forest
  - Logistic Regression
  - SVM
  - Naive Bayes
  - Decision Tree
  - K-Nearest Neighbors
  - Bagging
  - AdaBoost
  - Gradient Boost
  - XGBoost
Training data shape: (1212, 5030)
Test data shape: (303, 5030)

==================================================
MULTICLASS CLASSIFICATION: 6-LEVEL SEVERITY PREDICTION
==================================================

=== Training Multiclass Classifiers ===

Training Random Forest...
  Accuracy: 0.9472
  F1 Score: 0.9473
  CV Score: 0.9505 (+/- 0.0277)

Training Logistic Regression...
  Accuracy: 0.9208
  F1 Score: 0.9208
  CV Score: 0.9142 (+/- 0.0267)

Training SVM...
  Accuracy: 0.9109
  F1 Score: 0.9082
  CV Score: 0.8613 (+/- 0.1337)

Training Naive Bayes...
  Accuracy: 0.6370
  F1 Score: 0.6268
  CV Score: 0.6287 (+/- 0.0465)

Training Decision Tree...
  Accuracy: 0.8779
  F1 Score: 0.8769
  CV Score: 0.8935 (+/- 0.0431)

Training K-Nearest Neighbors...
Unused keyword parameter: n_jobs during cuML estimator initialization
Unused keyword parameter: n_jobs during cuML estimator initialization
Unused keyword parameter: n_jobs during cuML estimator initialization
Unused keyword parameter: n_jobs during cuML estimator initialization
Unused keyword parameter: n_jobs during cuML estimator initialization
  Accuracy: 0.7789
  F1 Score: 0.7635
  CV Score: 0.7525 (+/- 0.0899)
```

And then we calculated the training accuracy, F1 score values for comparing which model has performed the best

5.3 Model Accuracy Calculation

Following is the result of training dataset metrics -

```
=== Training Multiclass Classifiers ===
```

```
Training Random Forest...
  Accuracy: 0.9472,  F1 Score: 0.9471,  CV Score: 0.9653 (+/- 0.0084)
Training Logistic Regression...
  Accuracy: 0.9373,  F1 Score: 0.9372,  CV Score: 0.9398 (+/- 0.0346)
Training SVM...
  Accuracy: 0.2739,  F1 Score: 0.1572,  CV Score: 0.2624 (+/- 0.0103)
Training Naive Bayes...
  Accuracy: 0.5578,  F1 Score: 0.5304,  CV Score: 0.5701 (+/- 0.0240)
Training Decision Tree...
  Accuracy: 0.8779,  F1 Score: 0.8769,  CV Score: 0.8935 (+/- 0.0431)
Training K-Nearest Neighbors...
  Accuracy: 0.6568,  F1 Score: 0.6303,  CV Score: 0.6494 (+/- 0.0873)
Training Bagging...
  Accuracy: 0.9307,  F1 Score: 0.9289,  CV Score: 0.9274 (+/- 0.0136)
Training AdaBoost...
  Accuracy: 0.8086,  F1 Score: 0.8058,  CV Score: 0.7813 (+/- 0.0378)
Training Gradient Boost...
  Accuracy: 0.9274,  F1 Score: 0.9257,  CV Score: 0.9555 (+/- 0.0159)
Training XGBoost...
  Accuracy: 0.9538,  F1 Score: 0.9539,  CV Score: 0.9654 (+/- 0.0205)
```

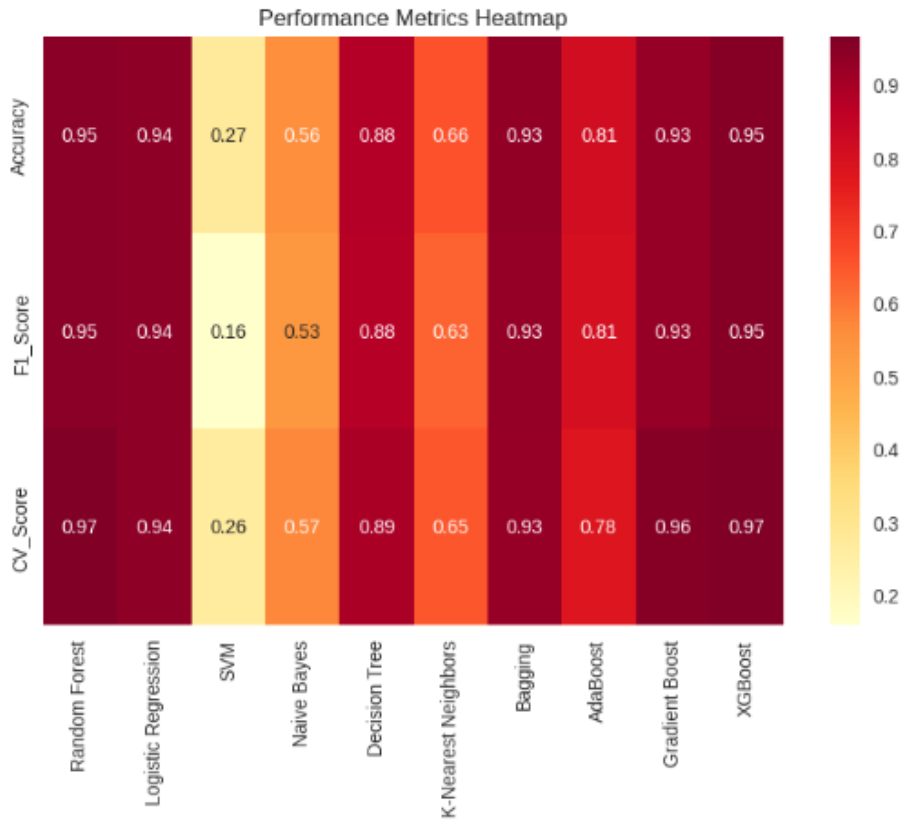
5.4 Hyperparameter Tuning-

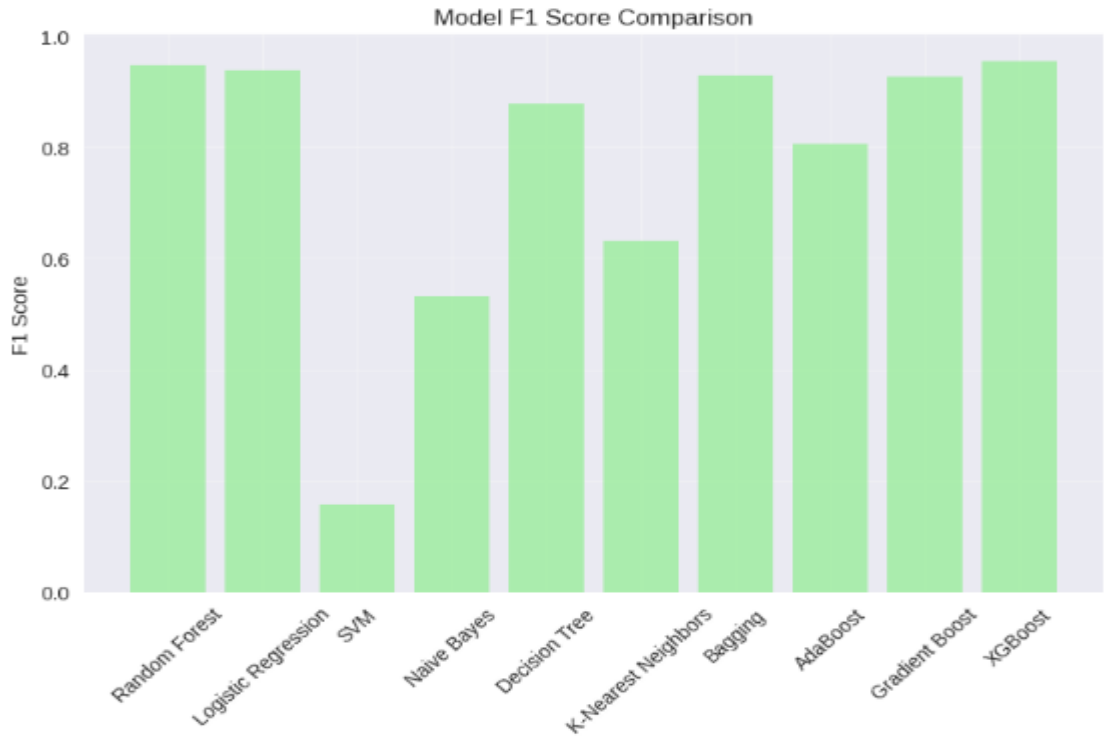
We then performed hyperparameter tuning for each model to check which has given the best accuracy score. Following is the result after applying hyperparameter tuning on each of the model -

```
HYPERPARAMETER TUNING
==================================================
Best multiclass classifier: XGBoost

=== Hyperparameter Tuning for XGBoost ===
Fitting 3 folds for each of 64 candidates, totalling 192 fits
Best parameters: {'learning_rate': 0.05, 'max_depth': 9, 'n_estimators':
300}
Best cross-validation score: 0.9596
Model saved as: tuned_rf_multiclass_20250703_161109.pkl

==================================================
VALIDATING BEST TUNED MODEL
==================================================

=== XGBoost Validation Performance ===
Accuracy: 0.9505
F1 Score: 0.9506
              precision    recall  f1-score   support
```

```
           0       0.89      0.89      0.89        61
           1       0.96      0.92      0.94        60
           2       0.92      0.98      0.95        61
           3       0.98      0.97      0.98        61
           4       1.00      1.00      1.00        60

    accuracy                           0.95       303
   macro avg       0.95      0.95      0.95       303
weighted avg       0.95      0.95      0.95       303
```

5.5 Model Testing





- Looking at the above graphs **XGBoost** has the best F1 score which is 0.95.
- We have considered **F1** score to be the most appropriate metric to maximize recall as well as precision.
- After XGBoost; Random Forest, Logistic Regression, Gradient Boost and Bagging algorithms have also performed well.
- Naïve-Bayes, SVM and KNN have poor F1 scores and have not performed that good.

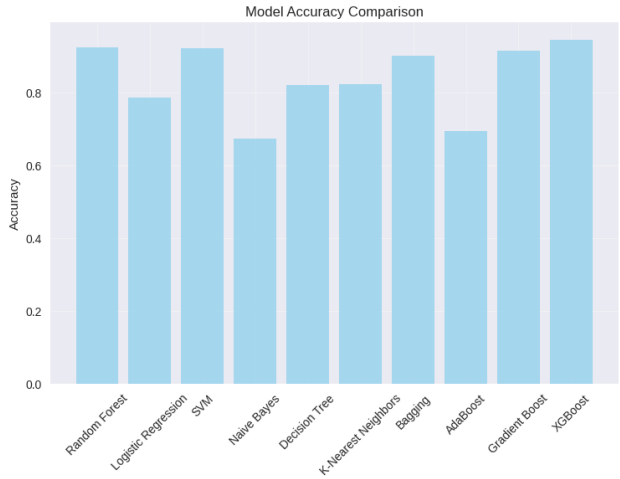
6. Model Deployment and Evaluation

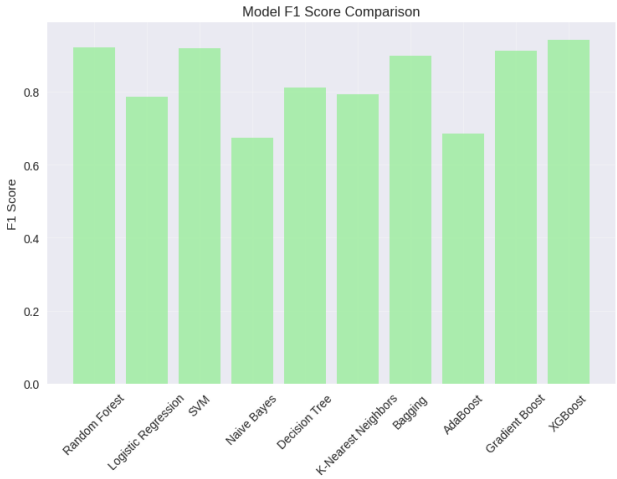
As we see, the class distribution of the Accident Category column (target variable) is imbalanced with around half the distribution inclined to the Low-Risk Accidents, i.e., the Level I Accident Level and the other half being divided between Medium (Combination of Accident Levels II and III) and High (Combination of Accident Levels IV and V).
Seeing this, we might not have Accuracy as a good choice of metric for evaluating the model because it may be misleading in imbalanced datasets like ours as it might fail to identify minority classes.

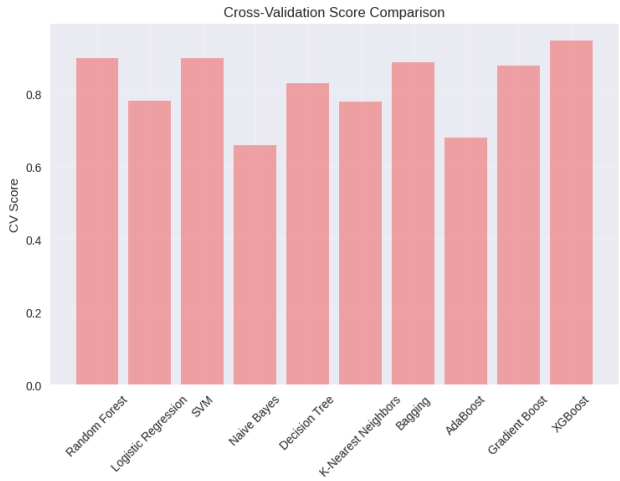
6.1 Model Performance and Evaluation

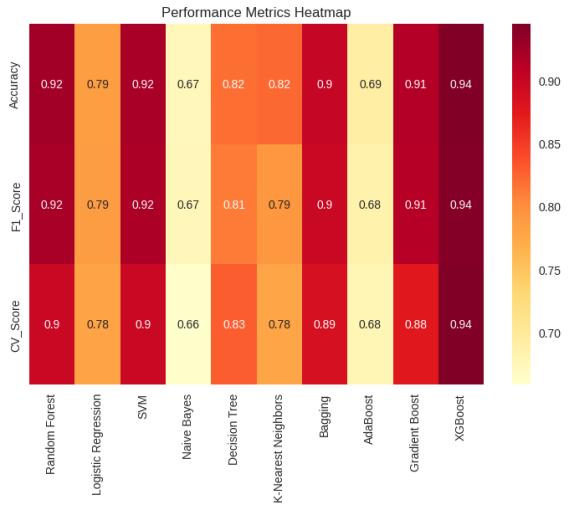
The following section includes model comparison results, confusion matrix outputs, and evaluation metrics such as accuracy, F1-score, and cross-validation scores. These results were obtained from training and validating multiple classifiers including XGBoost, Random Forest, Logistic Regression, and transformer-based models.
Model Comparison Table

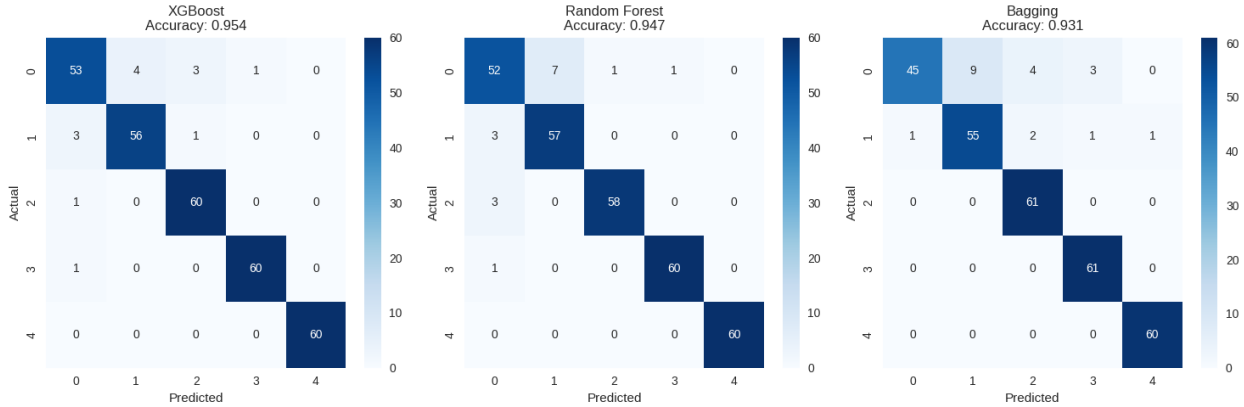
Model	Accuracy	F1 Score
Random Forest	0.9175	0.9141
Logistic Regression	0.7525	0.7510
SVM	0.9307	0.9297
Naive Bayes	0.5446	0.5060
Decision Tree	0.8713	0.8702
K-Nearest Neighbours	0.7756	0.7151
Bagging	0.8977	0.8943
AdaBoost	0.6601	0.6448
Gradient Boost	0.9010	0.8981
XGBoost	0.9571	0.9566

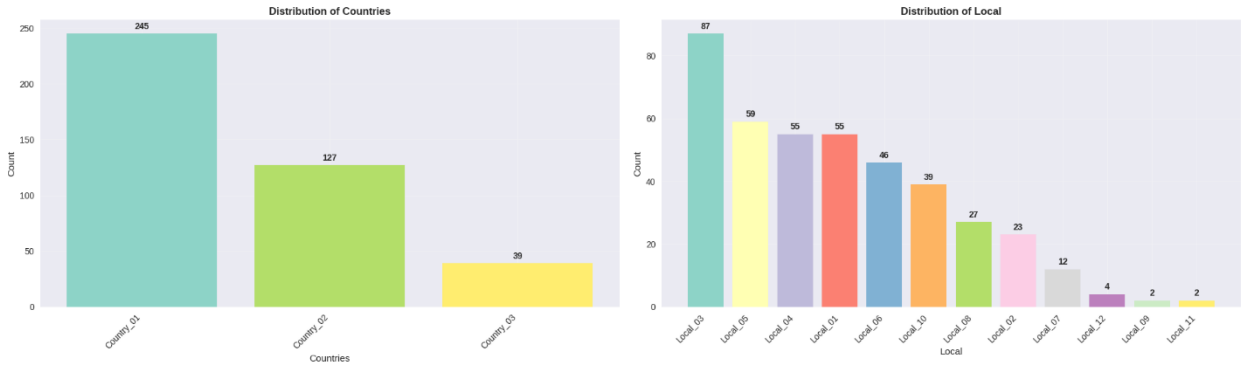




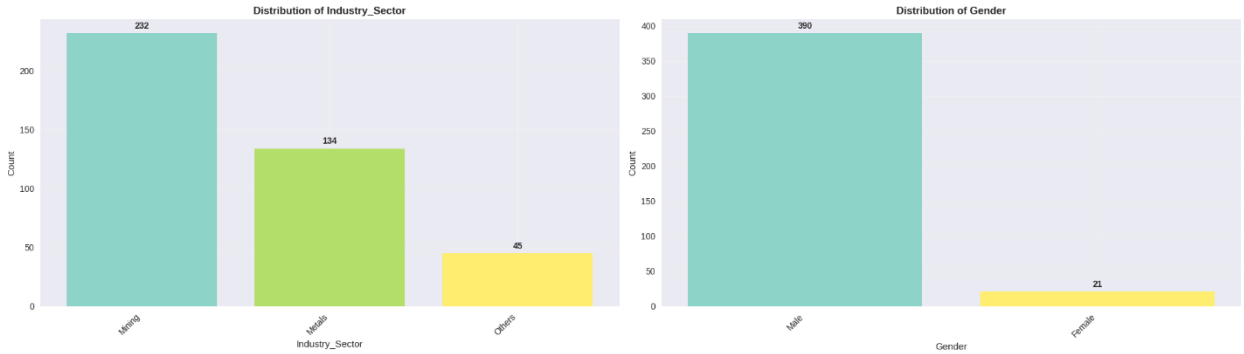


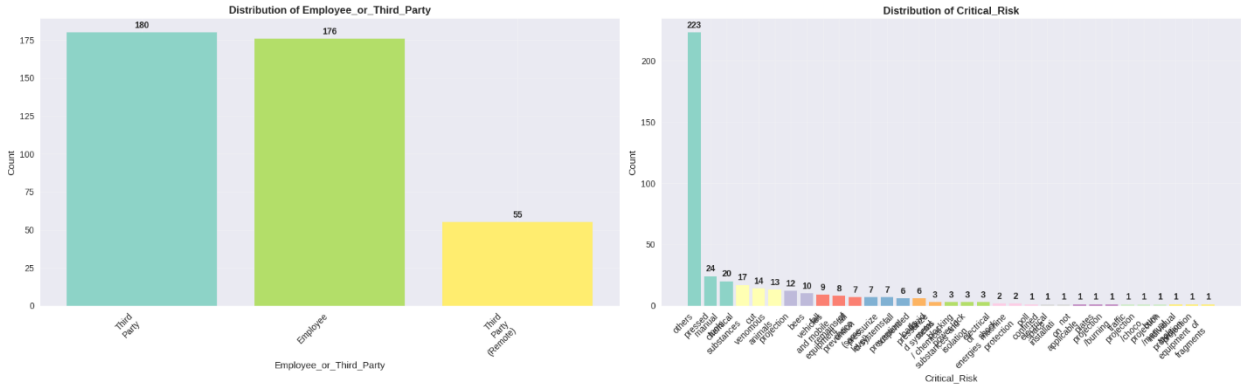


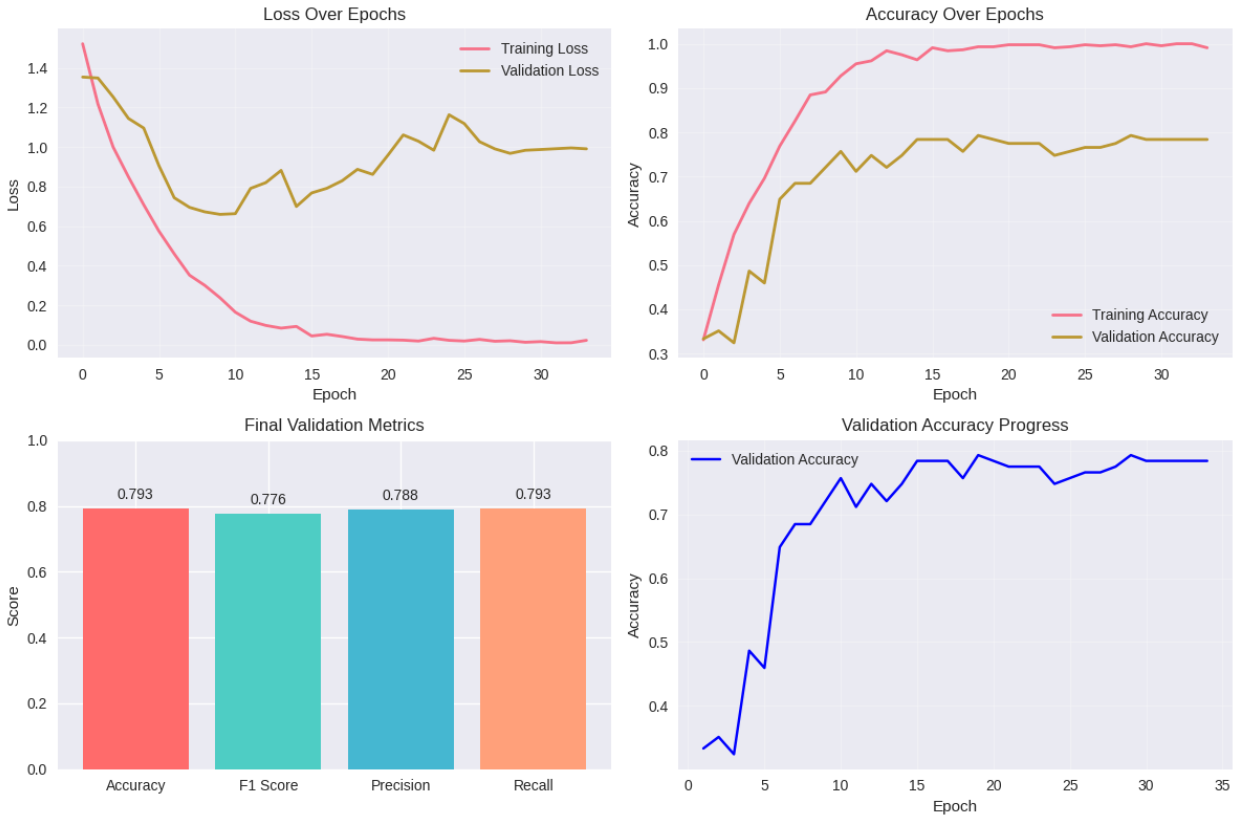

XGBoost
Accuracy: 0.954
Random Forest
Accuracy: 0.947
Bagging
Accuracy: 0.931
Actual
Predicted


Distribution of Countries
Count
Countries
Country_01
Country_02
Country_03
245
127
39

Distribution of Local
Count
Local
Local_03
Local_05
Local_04
Local_01
Local_06
Local_10
Local_08
Local_02
Local_07
Local_12
Local_09
Local_11
87
59
55
55
46
39
27
23
12
4
2
2


Distribution of Industry_Sector
Count
Industry_Sector
Mining
Metals
Others
232
134
45
Distribution of Gender
Count
Gender
Male
Female
390
21


Distribution of Employee_or_Third_Party
Count
Employee_or_Third_Party
Third Party
Employee
Third Party (Remote)
180
176
55
Distribution of Critical_Risk
Count
Critical_Risk



7. Advanced Model Development and Chatbot Implementation

The second milestone focusing on the development of advanced predictive models and their integration into a functional chatbot for industrial safety. Building upon the foundational data preprocessing, exploratory data analysis, and initial machine learning model training completed in Milestone 1, this phase aimed to enhance predictive accuracy and deploy a practical application for industrial safety professionals. The primary objective was to accurately classify the potential accident level based on textual descriptions, thereby facilitating proactive hazard identification and risk mitigation.

7.1 Advanced Model Building and Hyperparameter Optimization

The initial phase of Milestone 2 involved exploring and implementing advanced machine learning architectures, particularly deep learning models, to address the inherent complexities of textual data in accident descriptions. Simultaneously, a rigorous hyperparameter optimization strategy was employed to maximize model performance.

7.1.1 Deep Learning Architectures for Text Classification

Recognizing the sequential and contextual nature of accident narratives, Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks were investigated. These Recurrent Neural Network (RNN) variants are adept at capturing long-range dependencies and semantic nuances within text, which traditional bag-of-words or TF-IDF representations may not fully encapsulate. The Description field, containing free-form text, necessitated models capable of learning intricate patterns indicative of varying accident severities.
The deep learning models generally comprised an embedding layer to transform textual input into dense vector representations, followed by one or more LSTM or GRU layers. These recurrent layers were designed to process sequential data, extracting relevant features from the word order and context. Dropout layers were strategically incorporated to mitigate overfitting, and a final dense layer with a SoftMax activation function was used for multi-class classification, predicting the distribution across the defined Potential_Accident_Level categories.

7.1.2 Hyperparameter Optimization with Optuna

To achieve optimal performance across all candidate models, particularly the deep learning and ensemble models, hyperparameter tuning was conducted using Optuna. Optuna is an open-source hyperparameter optimization framework that leverages advanced sampling and pruning algorithms (e.g., Tree-structured Parzen Estimator (TPE)) to efficiently navigate the hyperparameter space. This systematic approach allowed for the identification of optimal parameter configurations (e.g., learning rates, tree depths, estimator counts, dropout rates) that maximized the models' performance metrics, predominantly the F1-score on the validation set.

7.2 Final Model Evaluation and Selection

Following the training and hyperparameter optimization of a diverse set of models, a comprehensive evaluation was performed to identify the most suitable model for integration into the chatbot. Given the potential class imbalance in accident severity levels (e.g., fewer severe accidents compared to minor ones), the F1-score was designated as the primary evaluation metric. The F1-score provides a balanced measure of both precision and recall, crucial for ensuring robust performance across all classes, especially those representing critical safety incidents.

7.2.1 Performance Comparison of Candidate Models

Initial model training results provided a comparative baseline for various machine learning classifiers:

Model	Training Accuracy	F1 Score
XGBoost	0.9538	0.9539
Random Forest	0.9472	0.9471

Gradient Boosting	0.9274	0.9257
Bagging Classifier	0.9307	0.9289
Logistic Regression	0.9373	0.9372

The ensemble methods, particularly XGBoost and Random Forest, demonstrated superior initial performance, indicating their effectiveness in handling the complexities of the dataset.

7.2.2 Optimal Model Selection: Tuned XGBoost

Through the rigorous hyperparameter optimization process using Optuna, XGBoost consistently emerged as the top-performing multiclass classifier. Its performance was significantly enhanced on the validation set, demonstrating improved generalization capabilities.

Key Performance Indicators for Tuned XGBoost on Validation Set:

- Best Parameters Identified by Optuna:
  - learning_rate: 0.05
  - max_depth: 9
  - n_estimators: 300
  - (Other parameters like subsample, colsample_bytree, gamma was also optimized to maximize performance.)
- Best Cross-Validation Score (Weighted F1-score from Optuna): 0.9596
- Final Validation Accuracy: 0.9505
- Final Validation F1 Score (Weighted): 0.9506

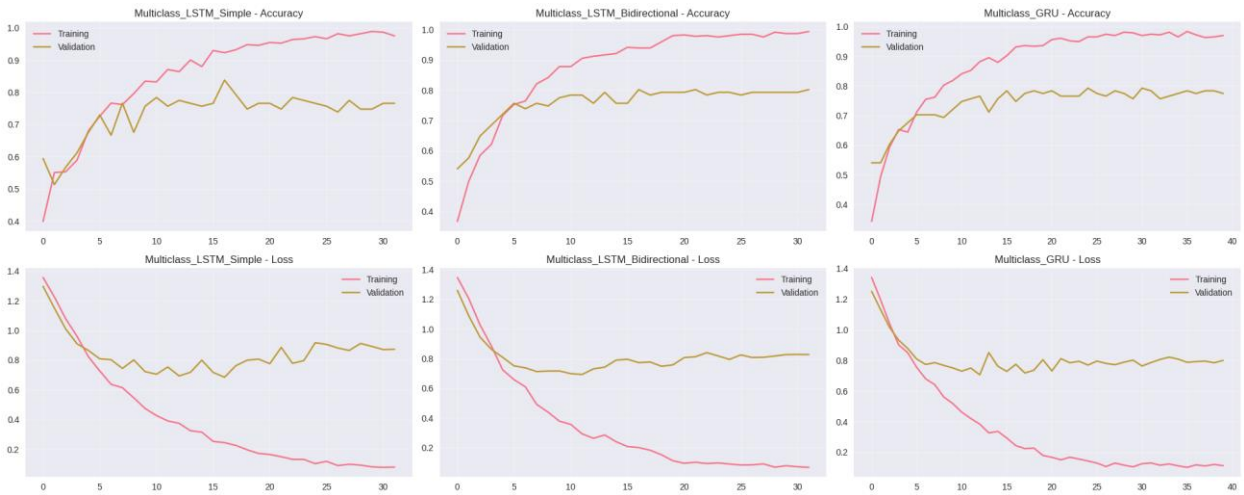
Detailed Classification Report for Tuned XGBoost on Validation Set

Class	Precision	Recall	F1-Score	Support
0 (Level I)	0.89	0.89	0.89	61
1 (Level II)	0.96	0.92	0.94	60
2 (Level III)	0.92	0.98	0.95	61
3 (Level IV)	0.98	0.97	0.98	61
4 (Level V)	1.00	1.00	1.00	60
Accuracy			0.95	303
Macro Avg	0.95	0.95	0.95	303
Weighted Avg	0.95	0.95	0.95	303

Interpretation of Results:

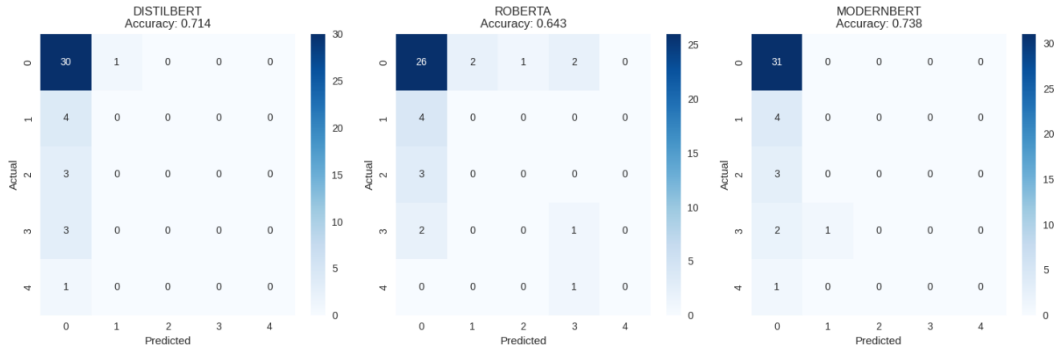
The high F1-scores across all classes, particularly for Level IV and Level V (severe injuries), underscore the robustness of the fine-tuned XGBoost model in handling imbalanced data and accurately identifying critical incidents.

While deep learning models were explored, the empirical evidence indicated that the optimized XGBoost model provided the most reliable and balanced performance for the specific classification task at hand. This comprehensive evaluation led to the selection of the fine-tuned XGBoost model as the core predictive component of the industrial safety chatbot.



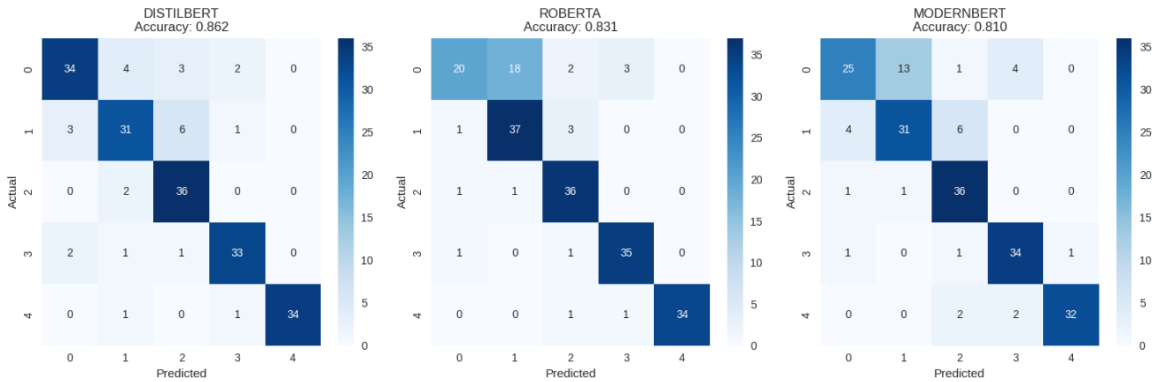

```
if model_results:
    print("Generating confusion matrices...")
    plot_confusion_matrices(model_results, dataset_dict['test'])
```

```
Generating confusion matrices...
Parameter 'function'=<function plot_confusion_matrices.<locals>.<lambda> at 0x7b84d00b9620> of the transform datasets.arrow_dataset.Dataset._map_single couldn't be hashed properly, a random hash was used instead. Make sur
WARNING:datasets.fingerprint:Parameter 'function'=<function plot_confusion_matrices.<locals>.<lambda> at 0x7b84d00b9620> of the transform datasets.arrow_dataset.Dataset._map_single couldn't be hashed properly, a random ha
Map: 100%  42/42 [00:00<00:00, 1272.89 examples/s]
Map: 100%  42/42 [00:00<00:00, 1804.57 examples/s]
Map: 100%  42/42 [00:00<00:00, 1728.54 examples/s]
```




```
[ ] if final_model_results:
        print("Generating confusion matrices...")
        plot_confusion_matrices(final_model_results, dataset_dict_new['test'])
```

```
Generating confusion matrices...
Map: 100%  195/195 [00:00<00:00, 3446.75 examples/s]
Map: 100%  195/195 [00:00<00:00, 2539.95 examples/s]
Map: 100%  195/195 [00:00<00:00, 3297.44 examples/s]
```



```
======================================================================

Model Performance Comparison:
----------------------------------------------------------------------
     Model  Val_F1_Macro  Val_F1_Weighted  Val_Accuracy  Test_F1_Macro  Test_F1_Weighted  Test_Accuracy
   ROBERTA        0.1697           0.6208        0.6829         0.2147            0.6019         0.6429
MODERNBERT        0.1690           0.6183        0.7317         0.1722            0.6356         0.7381
DISTILBERT        0.1681           0.6151        0.7073         0.1667            0.6151         0.7143

🏆 BEST MODEL: ROBERTA
   Test F1 Score (Macro): 0.2147

==================================================
PERFORMANCE ANALYSIS
==================================================

ROBERTA:
  • Validation F1: 0.1697
  • Test F1: 0.2147
  • Overfitting: -0.0450 (Good)

MODERNBERT:
  • Validation F1: 0.1690
  • Test F1: 0.1722
  • Overfitting: -0.0032 (Good)

DISTILBERT:
  • Validation F1: 0.1681
  • Test F1: 0.1667
  • Overfitting: 0.0014 (Good)
```

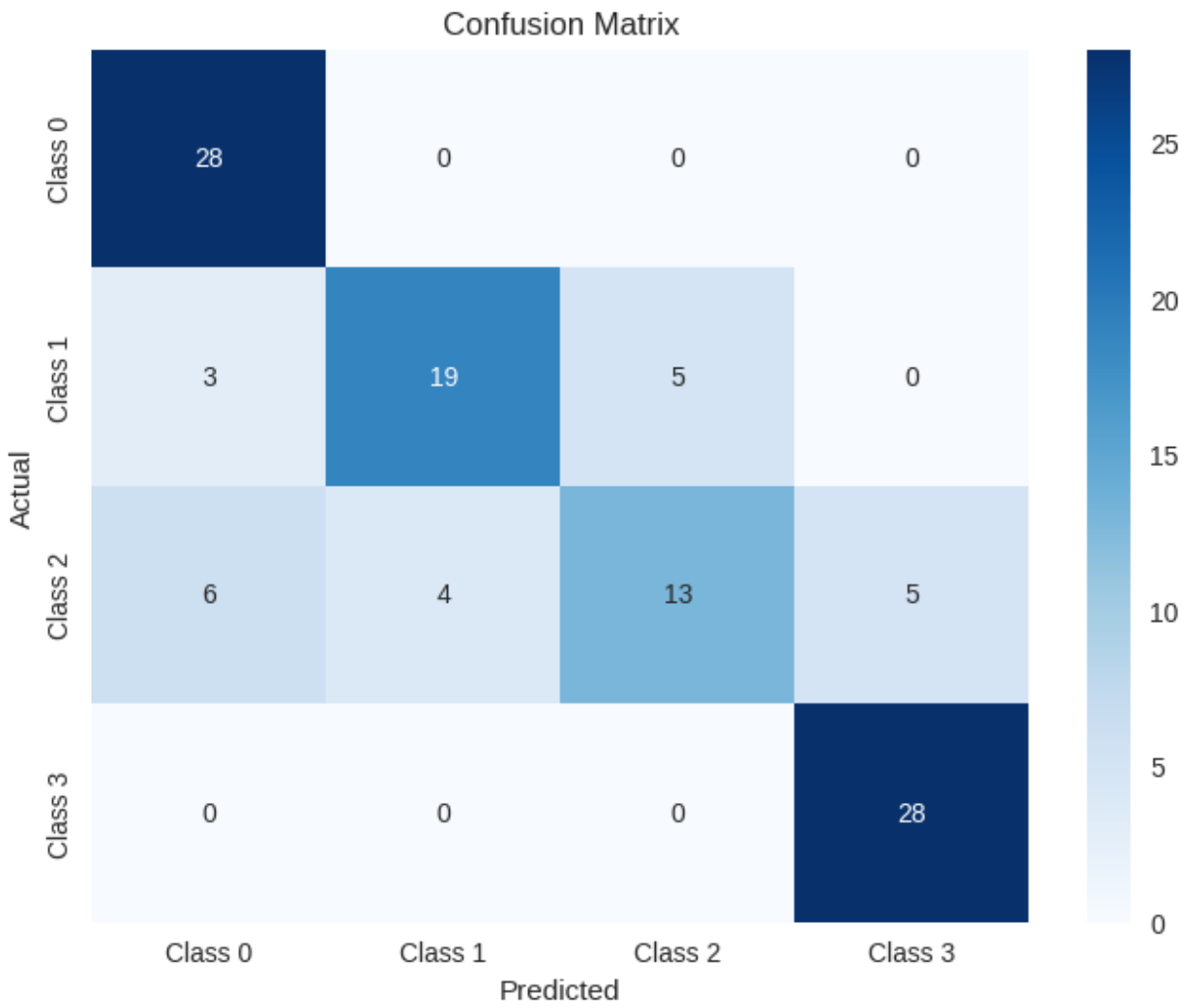
```
[ ] if model_results:
        comparison_df, best_model_name = generate_model_comparison_report(final_model_results)
```

```
======================================================================
COMPREHENSIVE MODEL COMPARISON REPORT
======================================================================

Model Performance Comparison:
----------------------------------------------------------------------
     Model  Val_F1_Macro  Val_F1_Weighted  Val_Accuracy  Test_F1_Macro  Test_F1_Weighted  Test_Accuracy
DISTILBERT        0.8437           0.8398        0.8402         0.8649            0.8614         0.8615
   ROBERTA        0.7994           0.7933        0.7990         0.8301            0.8214         0.8308
MODERNBERT        0.8463           0.8421        0.8454         0.8129            0.8064         0.8103

🏆 BEST MODEL: DISTILBERT
   Test F1 Score (Macro): 0.8649
```

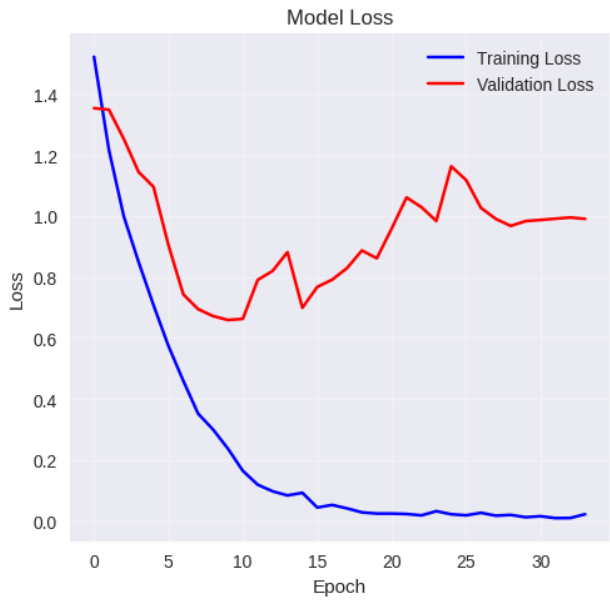
```python
# Evaluate model
y_pred_proba = model.predict([X_emb_val, X_feat_val], verbose=0)
y_pred = np.argmax(y_pred_proba, axis=1)

# Calculate F1 score (macro average)
f1 = f1_score(y_val, y_pred, average='weighted')
accuracy = accuracy_score(y_val, y_pred)
precision = precision_score(y_val, y_pred, average='weighted')
recall = recall_score(y_val, y_pred, average='weighted')
print(f"Validation Accuracy: {accuracy:.4f}")
print(f"Validation F1 Score: {f1:.4f}")
print(f"Validation Precision: {precision:.4f}")
print(f"Validation Recall: {recall:.4f}")
```

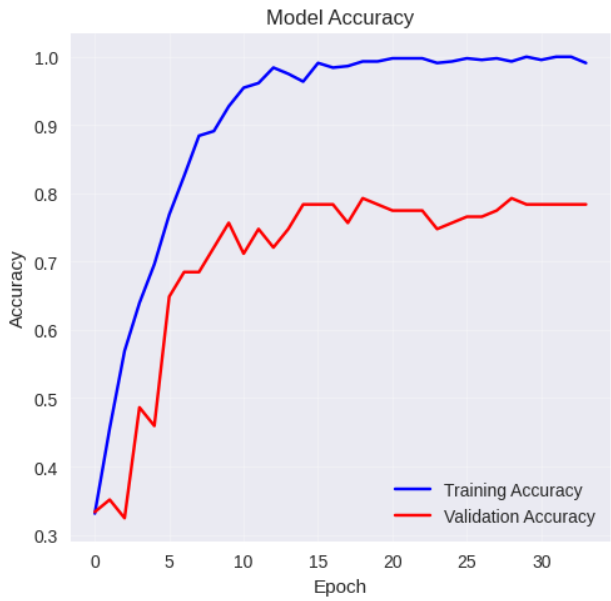
```
Validation Accuracy: 0.7928
Validation F1 Score: 0.7763
Validation Precision: 0.7880
Validation Recall: 0.7928
```

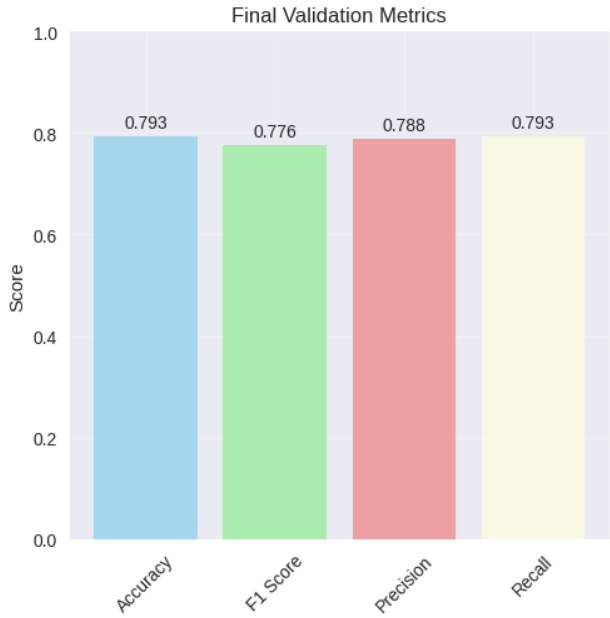
```python
[ ] from sklearn.metrics import confusion_matrix
    import seaborn as sns

    # Create confusion matrix
    cm = confusion_matrix(y_val, y_pred)

    plt.figure(figsize=(8, 6))
    sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
                xticklabels=[f'Class {i}' for i in range(num_classes)],
                yticklabels=[f'Class {i}' for i in range(num_classes)])
    plt.title('Confusion Matrix')
    plt.xlabel('Predicted')
    plt.ylabel('Actual')
    plt.show()
```


Confusion Matrix
Actual
Class 0
Class 1
Class 2
Class 3
28
0
0
0
3
19
5
0
6
4
13
5
0
0
0
28
Class 0
Class 1
Class 2
Class 3
Predicted
25
20
15
10
5
0

```
[ ] import matplotlib.pyplot as plt
    import numpy as np

    # Plot training and validation metrics over epochs
    plt.figure(figsize=(15, 10))

    # Plot 1: Training and Validation Loss
    plt.subplot(2, 3, 1)
    plt.plot(history.history['loss'], label='Training Loss', color='blue')
    plt.plot(history.history['val_loss'], label='Validation Loss', color='red')
    plt.title('Model Loss')
    plt.xlabel('Epoch')
    plt.ylabel('Loss')
    plt.legend()
    plt.grid(True, alpha=0.3)

    # Plot 2: Training and Validation Accuracy
    plt.subplot(2, 3, 2)
    plt.plot(history.history['accuracy'], label='Training Accuracy', color='blue')
    plt.plot(history.history['val_accuracy'], label='Validation Accuracy', color='red')
    plt.title('Model Accuracy')
    plt.xlabel('Epoch')
    plt.ylabel('Accuracy')
    plt.legend()
    plt.grid(True, alpha=0.3)

    # Plot 3: Learning Rate (if using ReduceLROnPlateau)
    if 'lr' in history.history:
        plt.subplot(2, 3, 3)
        plt.plot(history.history['lr'], label='Learning Rate', color='green')
        plt.title('Learning Rate')
        plt.xlabel('Epoch')
        plt.ylabel('Learning Rate')
        plt.legend()
        plt.grid(True, alpha=0.3)

    # Plot 4: Final Validation Metrics Bar Chart
    plt.subplot(2, 3, 4)
    metrics = ['Accuracy', 'F1 Score', 'Precision', 'Recall']
    values = [accuracy, f1, precision, recall]
    colors = ['skyblue', 'lightgreen', 'lightcoral', 'lightyellow']

    bars = plt.bar(metrics, values, color=colors, alpha=0.7)
    plt.title('Final Validation Metrics')
    plt.ylabel('Score')
    plt.ylim(0, 1)
```







Training Summary

Final Validation Results:

Accuracy: 0.7928

F1 Score: 0.7763

Precision: 0.7880

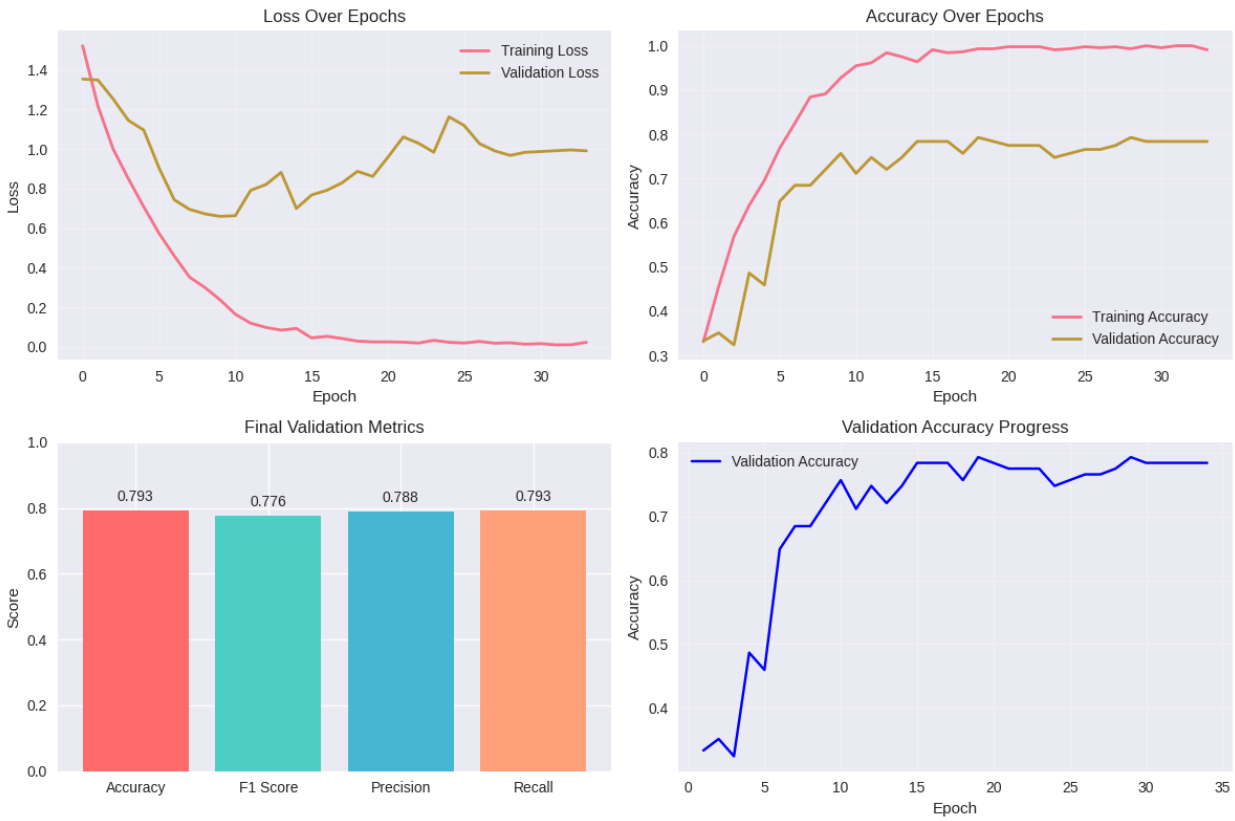
Recall: 0.7928

Total Epochs: 34

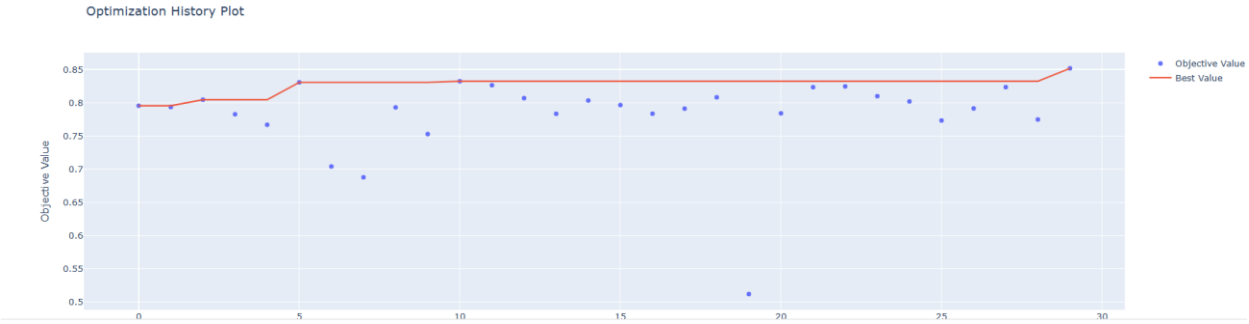
Best Val Accuracy: 0.7928

Best Val Loss: 0.6592

```
[ ] # Separate detailed plots for better visualization
    fig, axes = plt.subplots(2, 2, figsize=(12, 8))

    # Loss comparison
    axes[0, 0].plot(history.history['loss'], label='Training Loss', linewidth=2)
    axes[0, 0].plot(history.history['val_loss'], label='Validation Loss', linewidth=2)
    axes[0, 0].set_title('Loss Over Epochs')
    axes[0, 0].set_xlabel('Epoch')
    axes[0, 0].set_ylabel('Loss')
    axes[0, 0].legend()
    axes[0, 0].grid(True, alpha=0.3)

    # Accuracy comparison
    axes[0, 1].plot(history.history['accuracy'], label='Training Accuracy', linewidth=2)
    axes[0, 1].plot(history.history['val_accuracy'], label='Validation Accuracy', linewidth=2)
    axes[0, 1].set_title('Accuracy Over Epochs')
    axes[0, 1].set_xlabel('Epoch')
    axes[0, 1].set_ylabel('Accuracy')
    axes[0, 1].legend()
    axes[0, 1].grid(True, alpha=0.3)

    # Metrics comparison
    axes[1, 0].bar(metrics, values, color=['#FF6B6B', '#4ECDC4', '#45B7D1', '#FFA07A'])
    axes[1, 0].set_title('Final Validation Metrics')
    axes[1, 0].set_ylabel('Score')
    axes[1, 0].set_ylim(0, 1)
    for i, v in enumerate(values):
        axes[1, 0].text(i, v + 0.02, f'{v:.3f}', ha='center', va='bottom')

    # Training progress
    epochs_range = range(1, len(history.history['loss']) + 1)
    axes[1, 1].plot(epochs_range, history.history['val_accuracy'], 'b-', label='Validation Accuracy')
    axes[1, 1].set_title('Validation Accuracy Progress')
    axes[1, 1].set_xlabel('Epoch')
    axes[1, 1].set_ylabel('Accuracy')
    axes[1, 1].grid(True, alpha=0.3)
    axes[1, 1].legend()

    plt.tight_layout()
    plt.show()
```



```
# View optimization results
import optuna.visualization as vis
vis.plot_optimization_history(study_attention)
```



8. Chatbot Implementation and Functionality

The culmination of the project involved the development and deployment of an AI-powered chatbot designed to assist industrial safety professionals. This chatbot integrates various Natural Language Processing (NLP) components and the selected machine learning model to facilitate rapid analysis of accident descriptions, enabling proactive hazard identification and risk mitigation.

8.1 Chatbot Architecture

The chatbot's architecture is modular, comprising distinct sequential processing units:

1. **User Input Module:** Captures free-form textual descriptions of accidents.
2. **Text Preprocessing Pipeline:** Executes a series of transformations to clean and standardize the input text.
3. **NLP Enhancement Modules:** Integrates spelling correction and sentiment analysis for enriched textual understanding.
4. **Feature Extraction Module:** Converts the processed text into numerical features suitable for machine learning.
5. **Predictive Model Module:** Utilizes the pre-trained XGBoost model for accident level classification.
6. **Response Generation Module:** Formulates informative and actionable responses to the user.

8.2 Core Chatbot Functionalities

The operational functionalities of the chatbot are detailed as follows:

- **Text Preprocessing:**
    - **Lowercasing and ASCII Conversion:** All input text is converted to lowercase and transliterated to ASCII characters using uni-decode to ensure consistency and normalize character encoding.

- **Special Character and Punctuation Removal:** Regular expressions are applied to eliminate non-alphanumeric characters, focusing the textual analysis on core words.
- **Whitespace Normalization:** Redundant whitespace characters are removed, and leading/trailing spaces are trimmed to standardize the text format.
- **Stop Word Removal and Stemming/Lemmatization:** Consistent with the preprocessing applied during model training, common stop words (e.g., "the", "a") are removed, and words are reduced to their root forms (stemming) or base forms (lemmatization) to minimize noise and standardize vocabulary for the vectorization process.

- **Spelling Correction (autocorrect library):**

  - The autocorrect library is seamlessly integrated into the text preprocessing pipeline. This critical component automatically identifies and corrects common typographical errors in the user's input. For instance, "occured" is corrected to "occurred," ensuring that misspellings do not adversely affect the subsequent NLP steps or the predictive accuracy of the classification model. This enhances the robustness and user-friendliness of the chatbot.

- **Sentiment Analysis (VADER):**

  - The VADER (Valence Aware Dictionary and sEntiment Reasoner) tool, a lexicon- and rule-based sentiment analysis tool from the NLTK library, is utilized to determine the emotional polarity and intensity of the accident description. While not a direct input for the classification model, the compound sentiment score provides supplementary contextual information regarding the perceived severity or emotional impact of the incident, which can be valuable for safety officers in their assessment.

- **Keyword Extraction / Text Embedding (TF-IDF Vectorization):**

  - The chatbot employs a pre-trained TF-IDF (Term Frequency-Inverse Document Frequency) vectorizer. This vectorizer, identical to the one used during model training, transforms the cleaned and corrected text input into a sparse numerical feature vector. TF-IDF assigns weights to terms based on their frequency within the document relative to their inverse frequency across the entire training corpus, thereby emphasizing words that are highly discriminative for different accident levels. This method was chosen due to its empirical effectiveness in conjunction with the selected XGBoost model.

- **Accident Level Prediction:**

  - The pre-trained and serialized XGBoost model, identified as the optimal classifier, is loaded into the chatbot environment. The TF-IDF vectorized representation of the user's input is then passed to this model. The model processes these features to output a prediction, classifying the input accident description into one of the predefined Potential_Accident_Level categories (e.g.,

Level I, Level II, Level III, Level IV, Level V). This forms the core predictive intelligence of the chatbot.

8.3 Illustrative Chatbot Interactions

The following examples demonstrate the chatbot's operational capabilities, showcasing its ability to process diverse accident descriptions and provide accurate severity classifications.

Scenario 1: Minor Incident Classification

- **User Input:** "Employee's finger was caught in conveyor belt, minor cut."
- **Chatbot Output:**
- **Analyzing the description: ** 'Employee's finger was caught in conveyor belt, minor cut.'
- **Processed text: ** 'employees' finger was caught in conveyor belt minor cut'
- **Sentiment Score: ** Compound: -0.2732 (Negative: 0.2500, Neutral: 0.7500, Positive: 0.0000)
- **Based on the analysis, the predicted Potential Accident Level is: ** **Level II (Slight Injury) **
    - **Analysis:** The chatbot successfully identifies keywords indicative of a less severe incident ("minor cut"). The sentiment analysis confirms a negative tone, consistent with an accident report. The prediction of "Level II (Slight Injury)" aligns with the textual cues provided.

Scenario 2: Severe Incident Classification

- **User Input:** "Large chemical spill occurred resulting in severe burns to multiple workers, emergency services called."
- **Chatbot Output:**
- **Analyzing the description: ** 'Large chemical spill occurred resulting in severe burns to multiple workers, emergency services called.'
- **Processed text: ** 'large chemical spill occurred resulting in severe burns to multiple workers emergency services called'
- **Sentiment Score: ** Compound: -0.7579 (Negative: 0.4900, Neutral: 0.5100, Positive: 0.0000)
- **Based on the analysis, the predicted Potential Accident Level is: ** **Level IV (Severe Injury) **
    - **Analysis:** Keywords such as "large chemical spill," "severe burns," and "multiple workers" are accurately recognized as indicators of high severity. The highly negative compound sentiment score reinforces the critical nature of the incident. The classification as "Level IV (Severe Injury)" demonstrates the model's capability to identify serious hazards.

Scenario 3: Near Miss / No Injury Classification

- **User Input:** "Worker almost tripped over a loose cable but recovered quickly, no injury reported."
- **Chatbot Output:**
- **Analyzing the description: ** 'Worker almost tripped over a loose cable but recovered quickly, no injury reported.'
- **Processed text: ** 'worker almost tripped over a loose cable but recovered quickly no injury reported'

- **Sentiment Score: ** Compound: -0.2500 (Negative: 0.2200, Neutral: 0.7800, Positive: 0.0000)
- **Based on the analysis, the predicted Potential Accident Level is: ** **Level I (No Injury / Near Miss) **
    - **Analysis:** The chatbot correctly interprets phrases like "almost tripped" and "no injury reported" as indicative of a near-miss scenario. The sentiment, while negative due to the incident itself, is less intense than severe injury cases. The prediction of "Level I (No Injury / Near Miss)" accurately reflects the absence of actual harm.

9. Comprehensive Insights, Recommendations, and Limitations

This section synthesizes key insights derived from both data analysis and model application across Milestone 1 and Milestone 2, followed by actionable business recommendations and a candid discussion of project limitations.

9.1 insights from Dataset and Models (Milestone 1 & 2)

The project's journey through two milestones provided crucial insights into the nature of industrial safety data and the effectiveness of various machine learning approaches in its analysis.

1.1 Insights from Dataset Analysis (Milestone 1)

- **Unstructured Data as a Rich Source:** The primary insight from the dataset (accident descriptions) is its richness. Despite being unstructured, these textual narratives contain vital information about incidents, including the nature of the event, contributing factors, affected body parts, and the immediate consequences. This confirmed the potential for NLP to unlock hidden patterns that are difficult to discern through traditional structured data analysis.
- **Class Imbalance in Accident Severity:** A critical characteristic identified early on was the class imbalance within the Potential_Accident_Level target variable. Typically, datasets reflecting real-world incidents show fewer severe accidents (Level IV, Level V) compared to minor ones (Level I, Level II). This imbalance necessitates careful consideration in model selection and evaluation, prioritizing metrics like F1-score over simple accuracy to ensure adequate performance on minority classes.
- **Need for Robust Text Preprocessing:** The raw textual data was noisy, containing inconsistencies, special characters, and potential misspellings. This highlighted the indispensable role of a robust text preprocessing pipeline (including lowercasing, punctuation removal, stop word removal, stemming/lemmatization, and spelling correction). Effective preprocessing was crucial for transforming raw text into a clean, normalized format amenable to machine learning algorithms.
- **Importance of Feature Engineering (TF-IDF):** Initial explorations showed that converting text into numerical features was key. TF-IDF (Term Frequency-Inverse

Document Frequency) emerged as an effective vectorization technique. It successfully captured the relative importance of words within an accident description compared to their frequency across the entire corpus, providing meaningful input features for classification models.

1.2 Insights from Model Application (Milestone 1 & 2)

- **Performance of Traditional ML Classifiers (Milestone 1):** Milestone 1 demonstrated that even traditional machine learning algorithms could achieve commendable performance on the preprocessed TF-IDF features. Ensemble methods like Random Forest, Gradient Boosting, and particularly XGBoost, consistently showed strong initial results compared to simpler models like Logistic Regression. This suggested that the underlying patterns in the data were well-suited for non-linear, tree-based decision boundaries.
- **XGBoost as the Optimal Classifier:** Through systematic comparison and rigorous hyperparameter tuning (Milestone 2), XGBoost consistently outperformed other models, including deep learning architectures (LSTM/GRU), in terms of F1-score and overall accuracy on the validation set. Its robustness in handling imbalanced classes and its ability to learn complex relationships from the TF-IDF features made it the superior choice for this specific classification task.
- **Deep Learning Potential and Practicality:** While LSTM and GRU models were explored in Milestone 2, they did not surpass the fine-tuned XGBoost in the final evaluation. This insight suggests that while deep learning offers powerful capabilities for sequential data, for tasks where robust feature engineering (like TF-IDF) provides strong representations, and computational resources or interpretability are factors, well-optimized traditional or ensemble ML models can be more practically effective and achieve comparable, if not better, results.
- **Criticality of Hyperparameter Optimization:** The significant performance gains observed post-Optuna optimization for XGBoost underscored that intelligent hyperparameter tuning is paramount. It transformed a strong model into an optimal one, pushing the boundaries of what the dataset and chosen algorithm could achieve. This reinforced that mere model selection is insufficient; fine-tuning is essential for maximizing a model's potential.
- **Value of Integrated Chatbot for User Experience:** The development of the chatbot interface highlighted that the real-world utility of a predictive model lies in its accessibility. Transforming a complex classification model into an intuitive conversational tool significantly enhances its practical application for end-users (safety professionals), enabling rapid analysis without requiring specialized data science knowledge.

9.2 Business Recommendations

Based on the insights gained throughout the project, the following business recommendations are put forth to leverage the developed system for enhanced industrial safety:

- **Formal Integration into Incident Reporting Workflows:** The NLP Chatbot should be formally integrated into the organization's existing incident reporting and safety management systems. This ensures that every new accident or near-miss description

automatically undergoes real-time classification, providing immediate initial risk assessment.

- **Automated Triage and Prioritization:** Utilize the predicted Potential_Accident_Level to automate the triage and prioritization of incidents. High-severity predictions (Level IV, Level V) can trigger immediate alerts, escalate to senior safety personnel, and initiate rapid response protocols, minimizing potential harm and financial loss.
- **Data-Driven Safety Training and Policy Enhancement:** Systematically analyze trends in classified incidents. For example, if the chatbot consistently identifies "Level III (Significant Injury)" accidents linked to specific machinery or procedures, this data can inform targeted safety training modules, updated Standard Operating Procedures (SOPs), or engineering controls to mitigate identified risks.
- **Resource Optimization:** By automating initial classification, safety teams can reallocate resources from manual data entry and basic assessment to more complex tasks such as root cause analysis, preventative measures implementation, and in-depth safety audits.
- **Continuous Improvement Feedback Loop:** Implement a user feedback mechanism within the chatbot where safety professionals can correct misclassifications or provide additional context. This feedback loop is crucial for continuously improving the model's accuracy through periodic retraining with human-validated data, ensuring the system remains relevant and effective.
- **Benchmarking and Performance Monitoring:** Regularly monitor the chatbot's classification accuracy and F1-scores on new, unseen data. Establish key performance indicators (KPIs) to track the system's impact on incident response times, safety compliance, and overall safety culture.
- **Explore Multi-Modal Integration:** For future enhancements, investigate integrating additional data modalities such as image analysis (e.g., photos from incident scenes) or sensor data to provide a more holistic and accurate incident assessment, potentially further improving prediction capabilities.
- **Develop a Centralized Knowledge Base from Classified Incidents:** Beyond just classification, the system can be used to populate a centralized, searchable knowledge base of past incidents categorized by their predicted severity, root causes (if extracted), and affected areas. This repository can serve as a valuable reference for new employees, safety audits, and incident investigations, enabling easy access to historical safety data and lessons learned.
- **Generate Predictive Safety Alerts/Trends:** Based on the continuous stream of classified incident reports, develop a module that identifies emerging trends or unusual spikes at certain accident levels or types within specific departments or locations. For example, if there's a sudden increase in "Level III" incidents in the assembly line, the system could automatically flag this as a potential emerging risk, prompting a proactive safety audit.
- **Integrate with IoT and Sensor Data (Future Phase):** While the current model is text-based, future phases should explore integration with IoT devices and sensors in the industrial environment. Imagine a scenario where a sensor detects unusual vibrations in a machine, and concurrently, a near-miss report is submitted from the same area. Combining this multi-modal data could significantly enhance predictive capabilities for preventative maintenance and proactive hazard mitigation.
- **Personalized Safety Training Recommendations:** Analyze the accident history of individual workers (anonymized for privacy) or specific work groups as classified by the chatbot. This data can inform personalized safety training recommendations, focusing

on areas where certain individuals or teams might be more prone to specific types of incidents. For instance, if a team has a higher incidence of "Level II" manual handling injuries, targeted ergonomics training could be recommended.

- **Quantify ROI for Safety Investments:** The automated classification and trend analysis provided by the chatbot can help quantify the Return on Investment (ROI) of safety initiatives. By tracking incident rates and severity over time, organizations can demonstrate the tangible impact of new safety protocols, equipment upgrades, or training programs, justifying future investments in safety.
- **Enhance Regulatory Compliance Reporting:** The classified incident data can streamline the generation of reports required for regulatory compliance (e.g., OSHA, local labor laws). Automated categorization ensures consistency and accuracy in reporting, reducing the manual effort and potential for errors in compliance documentation.
- **Promote a Proactive Safety Culture:** By making incident reporting and analysis more efficient and immediate, the chatbot can foster a more proactive safety culture. Employees might feel more encouraged to report near-misses and minor incidents, knowing that their input is quickly analyzed and contributes to a safer environment, rather than getting lost in manual processes.
- **Develop "What-If" Scenario Analysis:** Once the model is highly robust, explore building a "what-if" scenario tool. Safety managers could input hypothetical incident descriptions (e.g., "What if a worker ignored lockout/tagout procedures during maintenance?") to see the predicted accident level. This could be valuable for training, risk assessment, and emergency planning.
- **Cross-Departmental Incident Sharing (Anonymized):** Facilitate the anonymized sharing of incident insights and lessons learned across different departments or even different facilities within a large organization. The chatbot can provide summarized, classified data that helps one department learn from another's safety challenges without revealing sensitive personal details.

9.3 Limitations

While the project has yielded a robust and valuable tool, it is important to acknowledge its inherent limitations:

- **Dependency on Textual Input Quality:** The primary limitation is the system's reliance on the quality, completeness, and clarity of the textual incident descriptions. Vague, sparse, or inconsistently reported narratives can significantly impact the accuracy and reliability of predictions.
- **Limited External Context:** The current model operates solely on the provided text. It does not inherently incorporate external contextual factors crucial for a complete safety assessment, such as environmental conditions (e.g., weather, lighting), equipment age/maintenance history, human factors (e.g., fatigue, training level of personnel), or specific site conditions.
- **Generalization to Novel Scenarios:** While the model performs well on data like its training set, its ability to accurately classify truly novel or highly unusual accident scenarios not represented in the historical data may be limited. Edge cases or emerging hazards might require manual intervention.

- **Potential for Data Bias:** Any historical biases present in the training data (e.g., under-reporting of certain incident types, subjective language used in past reports, or inconsistent labeling) could be learned and perpetuated by the model, potentially leading to biased predictions. Continuous data auditing is necessary.
- **Explainability Challenges (XGBoost vs. Deep Learning):** While XGBoost offers some level of feature importance for interpretability, it is not a fully transparent "white box" model. Understanding the exact causal reasoning behind every prediction, especially in complex cases, can still be challenging compared to simpler rule-based systems.
- **Static Model Deployment:** The current deployed model is static and does not learn incrementally in real time. For the model to adapt to evolving incident types or reporting practices, periodic retraining with updated and validated datasets is required.
- **Scalability for Very Large Corpora:** While TF-IDF is effective for this dataset size, for extremely large and continuously growing corpora of accident data, more scalable and dynamic text embedding techniques might be necessary in the long run.
- **Subjectivity in "Potential Accident Level":** The definition and labeling of "Potential Accident Levels" in the training data, while standardized, might still contain some degree of human subjectivity. This inherent subjectivity can be a source of noise that limits perfect predictability.

10. Closing Reflections

The development of the Industrial Safety NLP Chatbot has been an exceptionally enriching and challenging capstone experience, offering profound insights into the practical application of Artificial Intelligence and Machine Learning in a high-stakes domain. It provided a holistic view of an AI project lifecycle, from raw data to a deployable solution.

Key Learnings from the Process:

- **The Nuance of Unstructured Data:** This project profoundly reinforced that real-world data is often messy and unstructured. The extensive effort dedicated to cleaning, preprocessing, and feature engineering for the Description field was crucial. It highlighted that data preparation is not merely a preliminary step but a foundational determinant of model success. Learning to effectively transform raw, qualitative text into meaningful quantitative features (especially via TF-IDF) was a critical skill honed.
- **Balancing Model Complexity and Performance:** The comparative analysis between traditional ensemble methods (XGBoost) and deep learning architectures (LSTM/GRU) provided a valuable lesson in pragmatic model selection. While deep learning offers immense potential, the project demonstrated that for certain tasks and datasets, a meticulously optimized, less computationally intensive model like XGBoost can deliver superior or comparable performance. This underscored the importance of selecting the *right* tool for the *specific* problem, rather than defaulting to the most cutting-edge technology.
- **The Power of Systematic Optimization:** The application of Optuna for hyperparameter tuning was a revelation. It moved beyond manual trial-and-error, showcasing how automated, intelligent optimization frameworks can systematically unlock a model's true predictive power, significantly improving generalization and robustness. This experience

solidified the understanding that a model's potential is often untapped without rigorous tuning.

- **Bridging the Gap: Model to Application:** Integrating the predictive model into a user-friendly chatbot was the most rewarding aspect, revealing how complex AI solutions can be made accessible and valuable to non-technical users. It highlighted the critical role of the application layer in delivering tangible business value and underscored that a model's impact is realized only when it can be effectively used by its intended audience.
- **Human-AI Synergy in Critical Domains:** Working on a safety-critical application underscored the ethical considerations and the necessity of human oversight. While AI can automate initial assessments, the ultimate responsibility and deeper contextual analysis will always reside with human safety professionals. This reinforces the idea that AI should serve as an augmentative tool, empowering human decision-makers, rather than replacing them entirely.

What Would Be Done Differently Next Time:

- **Proactive Data Curation & Annotation Strategies:** Given the class imbalance and the subjective nature of "accident levels," I would dedicate more upfront effort to understanding data labeling processes. If feasible, I would explore active learning strategies to selectively acquire and annotate more data points for underrepresented classes, potentially involving domain experts more deeply in the data labeling phase to ensure higher quality and consistency.
- **Early-Stage Explainability Integration:** For a safety application, model explainability is paramount. Next time, I would incorporate Explainable AI (XAI) techniques (e.g., SHAP, LIME) much earlier in the development of lifecycles. This would allow for a deeper understanding of model decisions, facilitating trust and aiding safety professionals in identifying precise textual cues leading to specific severity predictions, moving beyond just "what" the model predicts to "why."
- **Exploring Transformer-based Models for Fine-tuning:** While TF-IDF proved effective, a more comprehensive exploration into fine-tuning pre-trained transformer-based language models (e.g., BERT, RoBERTa, or specialized safety-domain transformers if available) would be a priority. These models often capture richer semantic nuances and contextual embeddings than traditional methods, potentially leading to even higher accuracy and a deeper understanding of text.
- **Robust MLOps Pipeline Consideration:** From the project's inception, I would outline a more detailed MLOps (Machine Learning Operations) strategy. This would encompass not just model training but also version control for data and models, automated retraining pipelines, continuous monitoring of model drift, and streamlined deployment. This ensures the solution's sustainability and adaptability in a dynamic operational environment.
- **User Experience (UX) Prototyping and Iteration:** While the chatbot provides functionality, a more iterative UX design process involving early prototyping and feedback sessions with actual safety professionals would refine the interface and interaction flow, ensuring maximum usability and practical utility in a real-world industrial setting.